Development of A Secure ReSale Marketplace for Electronics Devices for SysTech World

Aononto Jahan Junnurain

ID# 21103030

A Practicum in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)

Department of Computer Science and Engineering
IUBAT School of Computer Science and Engineering
IUBAT—International University of Business Agriculture and Technology

Spring 2026

A Secure ReSale Marketplace for Electronics Devices.

Aononto Jahan Junnurain

A Practicum in the Partial Fulfillment of the Requirements for the Award of Bachelor of Computer Science and Engineering (BCSE)

The practicum has been examined and approved,

____

Prof. Dr. Utpal Kanti Das
Professor and Chair, Dept. of CSE
Dean, ISCSE

____

Shahinur Alam
Co-supervisor, Coordinator and Assistant Professor

____

Krishna Das
Supervisor and Assistant Professor

Department of Computer Science and Engineering
IUBAT School of Computer Science and Engineering
IUBAT—International University of Business Agriculture and Technology

Spring 2026

Letter of Transmittal

13 May 2026

The Chair

Practicum Defense Committee

Department of Computer Science and Engineering

IUBAT—International University of Business Agriculture and Technology

4 Embankment Drive Road, Sector 10, Uttara Model Town

Dhaka 1230, Bangladesh.

Subject: Letter of Transmittal.

Dear Sir,

With due respect, it is my privilege to present the report on my practicum "A Secure Resale Marketplace for Electronic Devices", which I have prepared in part fulfillment of the requirement for the degree of Bachelor of the Science in Computer Science and Engineering (BCSE) from IUBAT. Practicum I undertook at SysTech World and where I was responsible for the creation and development of a secure system that protected the introduction of safe dealing practices and authentication of users. The results achieved during this practice report are as follows.

I take this opportunity to express my sincerest thanks to the SysTech World management for their kind cooperation and support during the whole period of the practicum.

Yours sincerely,

____

Aononto Jahan Junnurain

21103030

Organization's Certificate

Not Get Yet

Student’s Declaration

I declare that I have written the practicum report “A Secure Resale Marketplace for Electronic Devices” submitted to the IUBAT in partial fulfillment of the requirements for the degree of Bachelor of Computer Science and Engineering (BCSE). I also declare that this is an original report and has not been submitted elsewhere for any degree, diploma or certificate. All the information which I have taken from others is clearly acknowledged and I have listed them in the references. The findings and solutions in this report are the result of my own work and hands-on experience during the practicum.

Aononto Jahan Junnurain
21103030

Supervisor’s Certification

This is to certify that the practicum report titled “A Secure Resale Marketplace for Electronic Devices” has been completed by Aononto Jahan Junnurain, ID: 21103030, at the International University of Business Agriculture and Technology (IUBAT) as a part of partial fulfillment of the requirements for the award of the Bachelor of Computer Science and Engineering (BCSE) program. This report has been prepared under my supervision.

The performance of Aononto Jahan Junnurain is appreciable in terms of attendance, communication, problem-solving abilities, and sense of responsibility. The project work and final report have been reviewed, and it is confirmed that the practicum requirements have been satisfactorily fulfilled. He is now eligible to appear for the practicum defense. I wish him success and excellence in future endeavors.

Krishna Das

Supervisor and Assistant Professor

Department of Computer Science and Engineering

IUBAT—International University of Business Agriculture and Technology

Abstract

Safe convenience of an authenticated resale marketplace is necessary in order to ensure that the transaction for used mobile phones is secure. Traditional resale has its share of risks including potential for fraud, lack of buyer and seller verification, and insecure payment systems. We propose in this paper “A Secure Resale Marketplace for Electronic Devices,” a web application that maintains trust, openness and efficiency in the used electronic goods market. Built with latest web technologies, the solution emphasizes secure authentication, data privacy and user friendly interface.

Everything about this project is to set a safe place for a secure place for users to conduct resale activities. Features include user registration and authentication, role based access, secure transaction processing and product information management. The platform has built-in validations that deter forging and increases reliability. The development process was incremental and the system design and usability were improved steadily. Testing results show, that the system achieves higher level of transaction security, reduces risk, and leads to better user satisfaction.

In a nutshell, the project demonstrates that a secure web- based marketplace can properly enhance the efficiency, trustworthiness, and reliability of the electronic device repurposing markets.

Acknowledgments

Above all, I praise and thank to the Almighty Allah for His infinite mercy and blessings without which I would not have been able to complete this practicum. I am profoundly grateful to my venerable supervisor Krishna Das for permitting me to do this practicum under his guidance. His unwavering support, priceless advice, motivation and constructive comments were of utmost importance while writing the report. I truly appreciate his patience and kind cooperation.

Also I would like to thank the CSE department, IUBAT for their cooperation and guidance. I am grateful to the Head of the department and all the esteemed faculty members for their motivation and inspiration throughout my studies.

In the end I want to thank my parents, family, friends and well-wishers who were giving me endless encouragement and moral support. And incase if I have missed someone in list I thank you all There is definitely no way that I can repay you for making this practicum successful, from the bottom of my heart Thank you!

Table of Contents

Letter of Transmittal iii
Organization's Certificate iv
Student's Declaration v
Supervisor's Certification vi
Abstract vii
Acknowledgments... viii
Table of Contents ix
List of Figures xiii
List of Tables xv
Chapter 1. 1
1.1 Background of the Study 2
1.2 Methodology 2
1.2.1 Primary Sources 3
1.2.2 Secondary Sources 3
1.3 Objectives 3
1.3.1 Broad Objective 4
1.3.2 Specific Objectives 4
1.4 Process Model 4
1.4.1 Reasons for Choosing the Agile Model: 5
1.5 Feasibility Study 6
1.5.1 Technical Feasibility 6
1.5.2 Economic Feasibility 6
1.5.3 Operational Feasibility 7
Chapter 2. 8
2.1 Organization Vision 9
2.2 Organization Mission 10
2.3 Organization Services 10
2.4 Organizational Structure 10

2.5 My Position in this Organization 11
2.6 Address of the Organization 12
Chapter 3. 13
3.1 Requirement Engineering 14
3.1.1 Requirement Elicitation 14
3.1.2 Requirement Analysis 14
3.1.3 Requirement Specification 15
3.1.4 Requirement Validation 16
3.2 System Requirements 16
3.2.1 Functional Requirements 16
3.2.2 Non-Functional Requirements 18
3.3 Use Case Diagram 20
Chapter 4. 21
4.1 Activity Diagram 22
4.1.1 Buyer Operational Activity Diagram 22
4.1.2 Seller Operational Activity Diagram 23
4.1.3 Admin Operational Activity Diagram 24
4.1.4 Admin Dispute Resolution Activity Diagram 25
4.1.5 Escrow Transaction Activity Diagram 26
4.1.6 Product Life Cycle Activity Diagram 27
4.2 Swimlane Diagram 28
4.2.1 ReSale Marketplace Swimlane Diagram 28
4.2.2 Negotiation & Escrow Lock Swimlane Diagram 29
4.2.3 Logistic & Order Processing Swimlane Diagram 30
4.2.4 Resolution and Settlement Swimlane Diagram 31
4.3 Class Diagram 32
4.4 Sequence Diagram 33
4.5 CRC Cards (Class-Responsibility-Collaborator) 34
Chapter 5 38
5.1 Risk Management 39
5.1.1 Stages of Risk Management 39

5.2 Risk Identification....................39
5.3 Risk Analysis....................40
5.4 Risk Planning....................41
5.5 RMMM (Risk Mitigation Management and Monitoring) Planning....................42
Chapter 6.....................50
6.1 Project planning and scheduling....................51
6.2 Function Point Estimation....................51
6.2.1. Functionality, Input, Output....................52
6.2.2. Identify Complexity (Data Functions)....................53
6.2.3 Unadjusted Function Point Contribution....................54
6.2.3 Unadjusted Function Point Contribution....................55
6.2.4 Calculation of Total Degree of Influence (TDI)....................56
6.2.6. Final Calculation....................57
6.3 Project Scheduling....................58
6.3.1 Development Phases....................58
6.3.2 Project Gantt Chart....................59
Chapter 7.....................60
7.1 DFD (Data Flow Diagram)....................61
7.1.1 DFD Level 0 – Context Diagram of the Resale Marketplace....................61
7.1.2 DFD Level 1: Functional Decomposition of System Processes....................62
7.1.3 DFD Level 2: Detailed Escrow & Financial Flow....................63
7.2 Entity Relationship (ER) Diagram....................64
7.3 Database Field Design....................65
7.4 Interface Design....................65
7.4.1 Home page....................66
7.4.2 Registration Page....................66
7.4.3 Login Page....................68
7.4.4 Admin Interfaces....................68
7.4.5 Buyer Interfaces....................70
7.4.6 Seller Interfaces....................71
7.4.7 Invoice....................73

Chapter 8. 74
8.1 Test Cases 75
8.1.1 Test Case 1 (User Registration) 75
8.1.2 Test Case 1 (User Login) 76
8.1.3 Test Case 1 (Product Management) 77
8.1.4 Test Case 1 (Chat & Negotiation) 78
8.1.5 Test Case 1 (Wallet & Escrow) 79
8.2 Test Summary Table 79
Chapter 9 80
9.1 Ethical Considerations 81
9.1.1 Data Privacy and Security 81
9.1.2. Intellectual Property 81
9.1.3. User Impact 81
9.1.4. Bias and fairness 82
9.1.5. Responsibility to Stakeholders 82
9.1.6. Professional Responsibility 82
9.2 Sustainability Through Software Engineering 82
9.2.1. Economical Sustainability 83
9.2.2. Social Sustainability 83
Chapter 10. 84
10.1 Conclusion 85
10.2 Limitations 85
10.3 Future Plans 86
References 87

List of Figures

Figure 1.1 Agile Process Model..5
Figure 2.1 Organizational Structure..11
Figure 2.2 Organizational Location Map...12
Figure 3.1 Use Case Diagram 20
Figure 4.1 Buyer Operational Activity Diagram 22
Figure 4.2 Seller Operational Activity Diagram..23
Figure 4.3 Admin Operational Activity Diagram 24
Figure 4.4 Admin Dispute Resolution Activity Diagram 25
Figure 4.5 Escrow Transaction Activity Diagram 26
Figure 4.6 Product Life Cycle Activity Diagram...27
Figure 4.7 ReSale Marketplace Swimlane Diagram..28
Figure 4.8 Negotiation & Escrow Lock Swimlane Diagram.............................29
Figure 4.9 Logistic & Order Processing Swimlane Diagram 30
Figure 4.10 Resolution and Settlement Swimlane Diagram...............................31
Figure 4.11 Class Diagram of Resale Marketplace 32
Figure 4.12 Sequence Diagram of ReSale Marketplace 33
Figure 6.1 Resale Marketplace Development Roadmap.....................................59
Figure 7.1 DFD Level 0 – Context Diagram 61
Figure 7.2 DFD Level 1 – Functional Decomposition 62
Figure 7.3 DFD Level 2 – Detailed Escrow & Financial Flow 63
Figure 7.4 ER Diagram for ReSale Marketplace..64
Figure 7.5 Database and User Table’s Overview 65
Figure 7.6 Home Page...66
Figure 7.7 Seller View of the Registration Page...66
Figure 7.7 User Login Page 66
Figure 7.8 Buyer View of the Registration Page 67

Figure 7.9 Login Page....68
Figure 7.10 Admin Dashboard....68
Figure 7.11 Platform Order History Page....69
Figure 7.12 Platform Wallet Page....69
Figure 7.13 Buyer Dashboard Page....70
Figure 7.14 Buyer Order History Page....70
Figure 7.15 Buyer Negotiation Page....71
Figure 7.16 Seller Dashboard....71
Figure 7.17 Seller Order History Page....72
Figure 7.18 Seller Negotiation Page....72
Figure 7.19 Invoice....73

List of Tables

Table 4.1 CRC of User....34
Table 4.2 CRC of Product Image....34
Table 4.3 CRC of Product....35
Table 4.4 CRC of Chat Message....35
Table 4.5 CRC of Chat Session....36
Table 4.6 CRC of Offer....36
Table 4.7 CRC of Review....37
Table 4.8 CRC of Wallet Transaction....37
Table 5.1 Risk Identification....39
Table 5.2 Risk Analysis....40
Table 5.3 Risk Planning....41
Table 5.4 RMMM Planning for Risk 1 (Financial Security)....43
Table 5.5 RMMM Planning for Risk 2 (Marketplace Fraud)....43
Table 5.6 RMMM Planning for Risk 3 (Unauthorized Access)....44
Table 5.7 RMMM Planning for Risk 4 (Real-time Latency)....45
Table 5.8 RMMM Planning for Risk 5 (Legal Compliance)....45
Table 5.9 RMMM Planning for Risk 5 (Dispute Bottlenecks)....46
Table 5.10 RMMM Planning for Risk 5 (Scope Creep)....47
Table 5.11 RMMM Planning for Risk 5 (Database Locking)....47
Table 5.12 RMMM Planning for Risk 5 (Data Corruption)....48
Table 5.13 RMMM Planning for Risk 5 (Data Privacy Violations)....49
Table 6.1 Functionality, Input and Output....52
Table 6.2 Identify Complexity of Data Function....53
Table 6.3 Identify Complexity of Transaction function....54
Table 6.4 Total Unadjusted Function Point (UFP)....55
Table 6.5 Calculation of Total Degree of Influence (TDI)....56
Table 8.1 Test Case 1....75
Table 8.2 Test Case 2....76

Table 8.3 Test Case 3..77
Table 8.4 Test Case 4..78
Table 8.5 Test Case 5..79
Table 8.6 Test Summary Table...79

Chapter 1.

Introduction

Trading and Selling Second Hand Electronic Devices has changed dramatically from the days of local classifieds to giant ecommerce marketplaces online. With the rising prices of technology and the shrinking life cycles of products, users are increasingly looking for trustworthy platforms where they can either buy used devices or sell their own.

The ReSale Marketplace for Electronic Devices is a unique outgrowth of the general ReSale Marketplace; it is a secured P2P (peer-to-peer) web platform which you can consider as an efficient way to conduct your trading. It combines live chat functionalities with high end financial security to create a solution that is both modern and reliable for anyone trading electronic devices online.

1.1 Background of the Study

The pace of technological development has been accelerating in recent years with resulting rapid obsolescence of electronic equipment including smartphones, laptops, and cameras, leading to an expanding secondary market for these electronics. This is attributed to consumers' desire to have better value for their money, as well as environmental consciousness regarding the reduction of electronic waste (e-waste).

Nevertheless, trust is a major challenge for the current peer-to-peer (P2P) marketplaces. Buyers are reluctant to pay upfront because of the risk of fraud, and sellers are reluctant to send product without payment. Also, the communication is sometimes ineffective and caused negotiations been postponed and users to be dissatisfied.

In this context, we propose a secure and interactive platform. It combines an escrow-based payment system to guarantee the security of the transaction and a real-time communication module with WebSocket technology to facilitate negotiations among users.

1.2 Methodology

The development of the ReSale Marketplace for Electronic Devices was grounded in a comprehensive research process aimed at identifying the key challenges associated with peer-to-peer (P2P) online transactions. Particular attention was given to critical issues such as trust deficits, fraud risks, and inefficient communication between buyers and sellers. To ensure the resulting system is both highly reliable and user-centric, data was systematically gathered and analyzed from two primary categories of information sources.

1.2.1 Primary Sources

Primary data was collected through hands-on interaction with the system to evaluate its core functionality, reliability, and user experience.

- Performed end-to-end testing of critical system elements, such as WebSocket communication, escrow transactions, and real-time UI updates
- Simulation of Buyer, Seller, and Admin actions to evaluate negotiation process and ease of use of the checkout
- Audited the database transaction log to confirm financial integrity for the processing of escrow and commission.
- Iterative testing and tuning to improve stability and performance of the system was conducted.

1.2.2 Secondary Sources

Secondary data was collected from existing literature and industry resources to support system design, technology selection, and best practices.

- Consulted with FastAPI, Python, WebSockets and JS official docs to develop real-time system features
- Explored existing P2P marketplaces (ebay, swappa) to get a feel for what is typical in the industry for dispute resolution, UI, and overall experience
- Researched web-based tutorials, tech blogs, and journal articles on escrow and safe transaction systems

1.3 Objectives

The main goals of the proposed system are divided into two categories: broad objectives and specific objectives. These objectives define both the overall purpose and the detailed functional targets of the system.

1.3.1 Broad Objective

The focus of this work is to design and implement a secure, transparent and quite interactive ReSale Marketplace for Electronic Devices. It enables trusted and reliable peer-to-peer (P2P) trading of used electronic gadgets.

It tries to solve crucial problems such as trust, insecurity of payment, and poor communication in the current marketplaces. By applying escrow-based payment scheme and providing real-time communication mechanisms, the system creates a secure and efficient digital trading platform where users can buy and sell electronic gadgets without any concern.

1.3.2 Specific Objectives

- Use WebSocket technology to build a real-time communication and negotiation system, providing buyers and sellers an instant interaction.
- Implement a secure escrow-based payment system that protects buyer funds until they receive and confirm the product.
- Design an automated inventory system to reflect product availability and status in real-time.
- Develop a full-featured admin panel to manage users, disputes, platform activities and commissions.

1.4 Process Model

In order to build the ReSale Market for Electronic Gadgets, the Agile Software Development Process Model was utilized. Agile is a repetitive, flexible methodology that centers around ongoing development, testing and enhancement with brief development cycles, termed sprints. This strategy is especially ideal for systems that require real-time communication and protect financial transaction.

Rather than trying to build everything at once, the workload was split up into well-defined modules such as user authentication, product management, real-time chat, escrow payments, and handling disputes. Each component was independently designed, coded, and tested, allowing the project to proceed smoothly and the system to maintain stability. Due to the modular nature, even complex facilities, such as communication based on WebSocket, and escrow transactions, were seamlessly integrated.

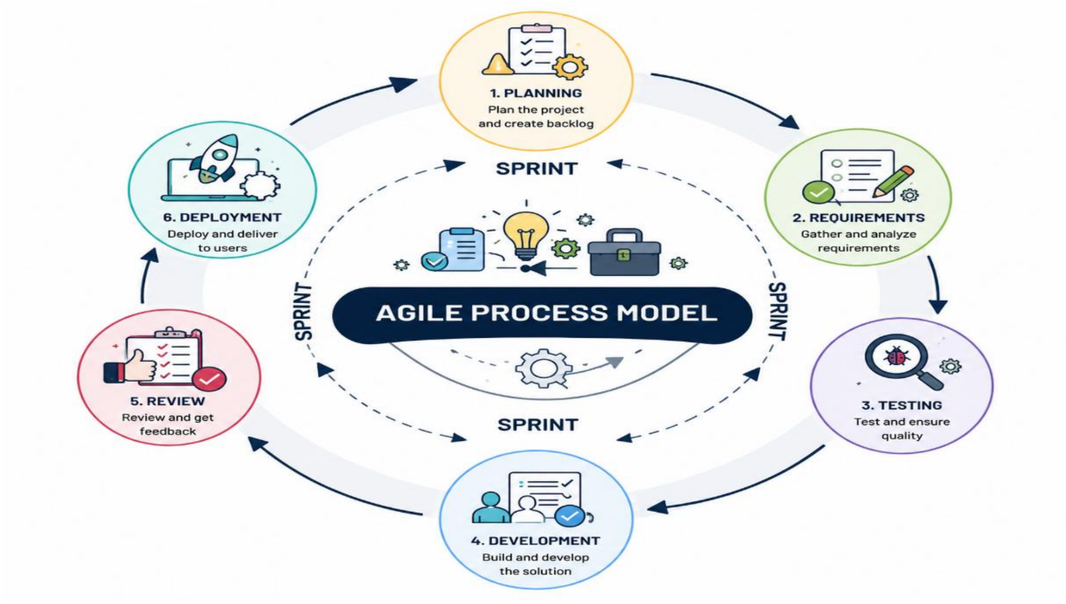


Figure 1.1 Agile Process Model

1.4.1 Reasons for Choosing the Agile Model:

- **Flexibility:** Agile made it possible for the system to evolve (for example, by improving negotiation procedures and escrow protocols) while maintaining its overall architecture.
- **Early Detection of Bugs:** Testing continuously at every stage led to early bug detection and fixing in the development cycle.
- **Incremental Delivery:** The application began with very basic business functions and progressed through more complex process such as on-line, real-time processing and secure payment.
- **User-Friendly Design:** Evaluation and testing was performed on a regular basis to increase usability and to provide a seamless experience for buyers and sellers alike.
- **Risk Mitigation:** Partitioning the high risk project into smaller modules led to mitigation of the risks and better control over the project.

1.5 Feasibility Study

Investigations of proposed system are categorized into three types: technical, economic and operational feasibility. It is an analysis to determine if the system can be developed and executed successfully within the existing resources, cost restraints and work environment. It also confirms that the system is realizable and that the cost and use of real-world system are justified.

1.5.1 Technical Feasibility

The proposed system can be realized with the use of modern and dependable technologies. Backend: 1 we are using FastAPI (Python) for the backend, which is a modern, fast (high-performance) web framework for building APIs with Python 3.10 +. With SQLAlchemy, database management is efficient and secure.

The frontend is open and built on html/css/JavaScript allowing the customized user to load the site on any browser for a lightweight, responsive experience. Also, WebSocket is added to allow real-time interaction among users with less latency and no more page refreshing. These combined technologies provide a system which is robust, scalable and well suited for both real time and subsequent market place trading.

1.5.2 Economic Feasibility

The economic feasibility is based on employing open-source technologies, which reduce drastically the development and software licensing costs. This enables the development of the system with very little upfront cost.

In addition, the platform has a revenue model which is based on a small percentage fee (0.5%) on successful trades. This means that any running costs, including server hosting, maintenance and software updates would be able to be covered as the site grew. In general, the system offers a inexpensive solution with a high chance of achieving sustainable financial viability in the long run.

1.5.3 Operational Feasibility

The suggested system is practically implementable because it is simple for the user to operate and the process is the same as the activities in the real-world marketplace. Sellers may list products with details and images, buyers and sellers negotiate prices via an integrated realtime chat system.

The administrators have a neat and functional dashboard to administer users, to keep an eye on transactions and to handle disputes. Furthermore, system-automated processes such as notifications and time-based escrow releases contribute to minimize manual interventions. The workflow of the system resembles real life BE processes and is therefore very easy to understand, adopt and use on a daily basis.

Chapter 2.

Organizational Overview

Systech World is a technology enabled aggressive growing company which provide best IT and BPO solutions in Dhaka, Bangladesh to various sectors. The company was founded to enable digital transformation with secure, scalable, and intelligent technology services.

Systech World has a wide range of solution: including IPPBX system, IVR system, CRM solution and call center software, Cloud Infrastructure services, AI powered automation, Tailor-made technical support outsourcing. Their clients range from telecommunications, to education, to healthcare, to Retail, to Government; showing adaptability and depth of expertise across industries.

It is backed by an energetic team of engineers, developers and technology professionals that collectively contribute to provide robust, effective technological solutions. Employing best practices and tools, with an ever evolving mind set towards quality, is how Systech World delivers constantly high-quality results.

In a nutshell, Systech World is a one-stop shop for everything tech! Systech World Values Innovation. Systech World regularly seeks out the most efficient solutions to keep their clients applications assured and their business running. With its high regard for creativity, honesty, and significance, Systech World strives to assist organizations to improve their operational effectiveness and compete in a progressively digital age.

2.1 Organization Vision

Systech World has a vision to lead as a technology solutions provider that is innovative, secure, and scalable by leading the digital transformation in various fields. The company strives to enable enterprises to maximize their business potential with cutting-edge IT & BPO solutions that improve business efficiency, productivity and long term growth.

Systech World is dedicated to inspiring innovation, being transparent and driving real change through smart technology that enables organizations to compete in an ever-evolving digital world."

2.2 Organization Mission

Systech World is a leading provider of secure, scalable and intelligent IT and BPO services that addresses the growing demand of today's enterprises. The company is dedicated to providing professional grade services with trained professionals, state of the art technology and industry best practices.

To uphold Systech World excellence through innovation, to maintain high ethical standards and to deliver "customer centric solutions" which positively impact the customer's operational performance. The company's objective is to help businesses realize digital transformation and deliver long-term value and sustainability.

2.3 Organization Services

Systech World provides a range of technology-based services aimed at supporting modern business operations and digital transformation. The key services offered by the organization include:

- Call center solutions such as IPPBX, IVR, and communication platforms with integration to CRM
- Custom development services with focus on web and desktop applications
- Business Process Outsourcing: customer support and back office services
- IT Professional Services including hardware/software support, maintenance and helpdesk.

2.4 Organizational Structure

The Systech World's organizational structure is based on hierarchal and function model to make certain that there is a very effective management and well-defined communication and coordination among the various divisions. The organization is designed with top management responsible for strategic decisions, followed by functional directors for the operational areas of software development, technical support, and business process outsourcing (BPO).

Teams within Departments each department is made up of various teams including developers, engineers, customer support, and operations staff that collaborate to provide services and solutions. This methodology allows for clear responsibilities and accountability within the business and most importantly a smooth flow of work throughout the organization.

The company's structure is designed to promote good decision-making in a way that enables us to deliver a consistently high level of service and respond to the changing needs of our business and the technology."

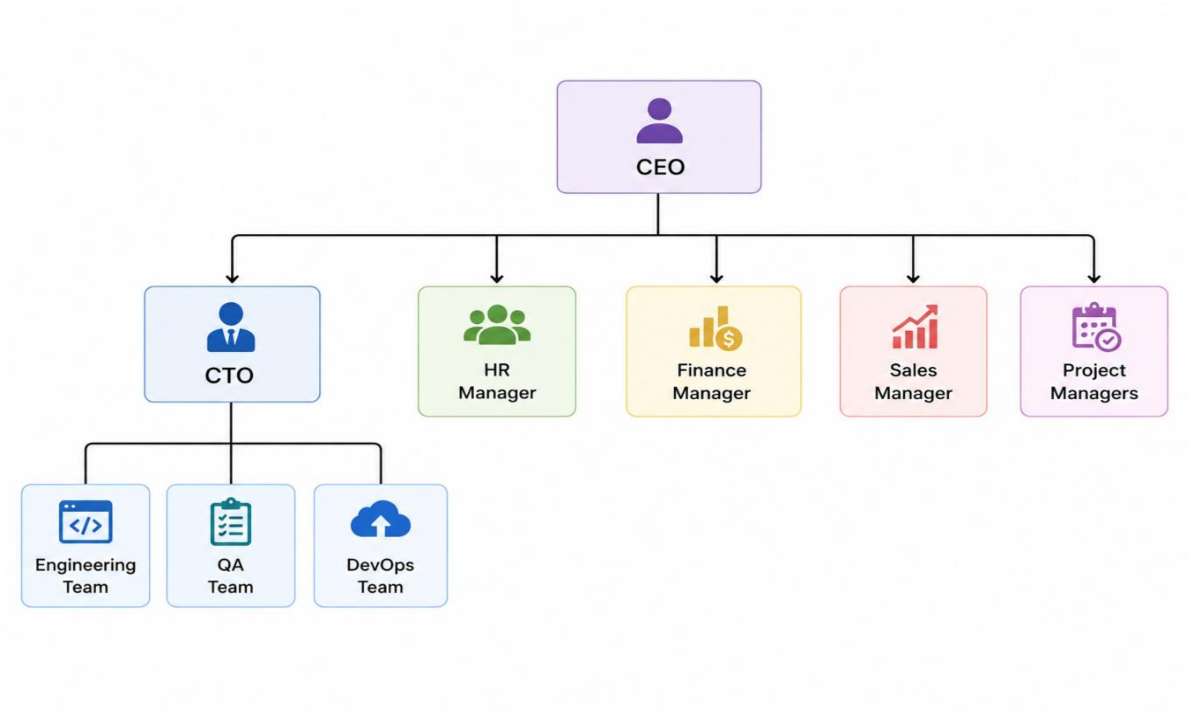


Figure 2.1 Organizational Structure

2.5 My Position in this Organization

I am working as an Intern Software Engineer at Systech World for the period of 6 Months. And I was responsible for assisting the software development team with the design, development, and maintenance of web applications.

Dungeon though I spent countless hours under guidance helping with both the frontend and backend developing implementing features, bugging issues and testing app functionality. Real Time Systems, APIs, and Database. I also obtained practical experience in Real Time systems, APIs and Database work under guidance of senior developers.

I self-learned a lot from previous experiences but not what I really call good quality real software development with source control, structured coding, sysob workflow and etc. This gave me an opportunity to relate both theoretical knowledge from the academics with practical projects, and at the same time hone my skills in problem solving, team work, and professional communications in a corporate setup.

2.6 Address of the Organization

The head office of Systech World is located at:

177/A, Dewanpara, Vashantek
Dhaka Cantonment, Dhaka
Bangladesh

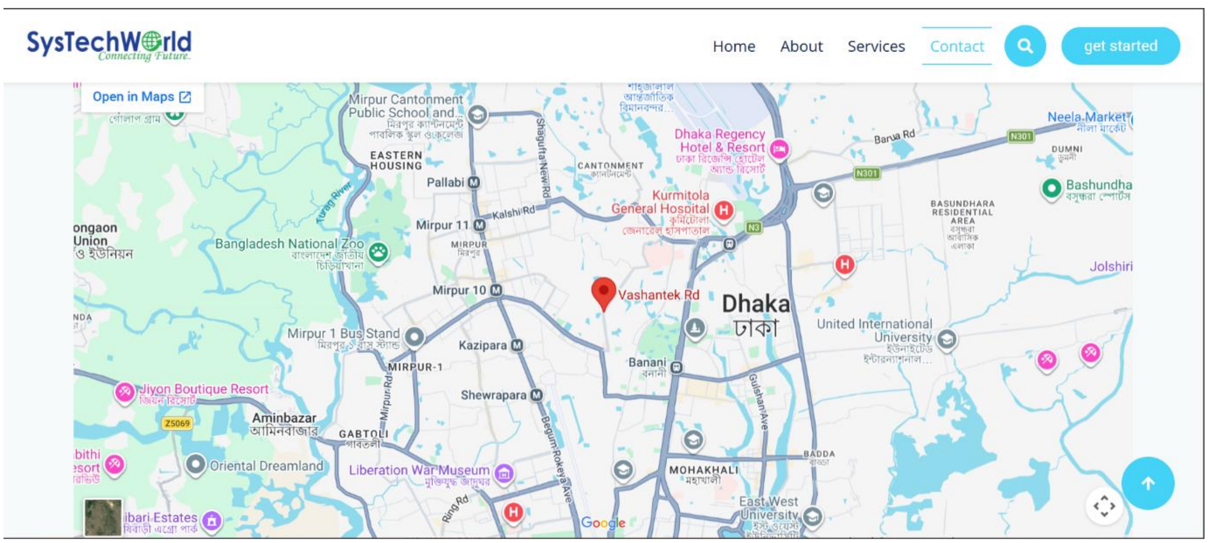


Figure 2.2 Organizational Location Map

Chapter 3.

Requirement Engineering

3.1 Requirement Engineering

Requirement Engineering is the command or orderly process of eliciting, analyzing, specifying and validating the requirements of ReSale Marketplace for Electronic Devices. This process is concentrated on gathering requirements for the system from primary stakeholders Buyers, Sellers, Administrators to address issues of trust, secure transactions, and real- time communication.

The methodology was implemented by applying a systematic procedure that specified both functional and nonfunctional requirements to the system and the aims of system development.

3.1.1 Requirement Elicitation

This phase involved collecting preliminary requirements from various stakeholders in order to gain insight into the desired functionalities and behaviors of the system. User and System Expectations. ReSale Marketplace for Electronics Devices, It was important to have a clear view of what the users wanted and system requirements. The following techniques were employed:

- Stake holder Analysis: Key users of the system, Buyers, Sellers and the Administrators and their roles, permissions, transaction workflow and system usage have been identified
- Observation: A study of established p2p marketplaces (e.g. Facebook Marketplace and the like) to find shared problems like no holding transaction security, no good communication tools, risky to face fraud…
- Similar Platforms Analysis: Examination of platforms like eBay and Swap to gauge typical functionalities and potential areas for enhancement, notably the incorporation of real-time chat-based negotiation and a secure escrow payment system

3.1.2 Requirement Analysis

After collecting the requirements, they were progressively analyzed for conflicts, ambiguities, and unrealism. In this way, it was guaranteed that all the requirements were unambiguous, attainable and consistent with the system goals.

- **Classification:** This is described in terms of the following Functional Requirements (the system shall perform escrow fund transactions, the system shall provide product listings, etc.) and Non-functional Requirements (the real-time chat system shall achieve low-latency message delivery over WebSocket technology, etc.).
- **Prioritization:** To build a secure, functional Minimum Viable Product (MVP), core features user authentication, product listing management, escrow wallet functionality, real-time communication, etc. were prioritized over ancillary ones.
- **Resolution of Conflicts:** Conflicting interests such as the seller's desire of getting paid immediately and the buyer's need to verify the product are resolved by means a transaction system based on secure escrow that retains the funds until delivery and confirmation.

3.1.3 Requirement Specification

Following the analysis, the needs and constraints were formally specified to describe system behavior and functionality in a precise manner.

- **System Specification:** A high-level overview of how the backend made with FastAPI (Python) handles the core logic, including database models, handling of the WebSocket connections, authentication, and escrow transactions. The front-end was developed in HTML, CSS and JavaScript, to be a dynamic-user-experience page using exposure of the data and on-line interactions of users.
- **Workflow definition:** The main system workflows have been (succinctly) tailored for operation. These are the Negotiation Cycle (Chat → Offer Made → Offer Accepted → Checkout) and the Order Cycle (Payment Held → Processing → Shipped → Delivered → Funds Released), with the purpose of transparency and uniformity throughout the transaction.

3.1.4 Requirement Validation

The final stage involved ensuring that the documented requirements accurately reflect stakeholder needs and system objectives.

- **Module Verification:** Key modules such as Buyer Dashboard, Seller Storefront, Admin Panel, and Live Chat Interface were reviewed for correctness and completeness
- **Goal Alignment:** System features were validated to ensure they support secure transactions, fraud prevention, and real-time communication
- **Feasibility Check:** Requirements were confirmed to be technically achievable with the chosen technologies

This step ensured that the system is accurate, practical, and aligned with user expectations.

3.2 System Requirements

This is the ReSale Market for ELECTRONIC DEVICES only, and includes the limits on ownership and operational restrictions, and system behavior restrictions. The functional perspective of the requirements can also be categorized into functional requirements describing what does a system do and non-functional requirements describing how does a system do.

3.2.1 Functional Requirements

These requirements define the specific functions and interactions within the system for different user roles.

- **User Authentication & Authorization**
  - The system allows users to securely register and log in to the platform.
  - The system must restrict access and route visibility based on user roles: Buyer, Seller, and Admin.
  - The system must require Admin approval for new Seller registrations before they can list products.

- **Admin Module**
  - User Management: Admins can view all registered users and apply disciplinary actions (e.g., 7-day bans, permanent account suspensions, or seller approvals).
  - Dispute Management: Admins have the authority to review frozen escrow transactions and issue partial refunds, full refunds, or forced seller payouts based on their investigation.
  - Platform Analytics: Admins can view real-time platform statistics, including total users, active listings, and platform commission revenue.
- **Seller Module**
  - Product Management: Sellers can upload new electronic devices, set prices, define inventory quantities, and upload up to 5 product images.
  - Order Fulfillment: Sellers can update the status of active orders (Processing, Shipped, and Delivered) and input courier tracking information.
  - Wallet Dashboard: Sellers can view their current wallet balance, historical payouts, and recent platform commission deductions.
- **Buyer Module**
  - Marketplace Browsing: Buyers can search, filter (by category), and view public listings of approved electronics.
  - Negotiation: Buyers can utilize the real-time chat interface to send custom financial offers to sellers before checking out.
  - Order Tracking & Feedback: Buyers can track their purchases, release escrow funds upon safe delivery, and leave verified 1-to-5 star reviews on the seller's profile.

- **Escrow & Wallet Module**
  - Fund Holding: The system must automatically deduct funds from the buyer's wallet and hold them in a secure Escrow state upon purchase.
  - Auto-Release: The system must automatically release escrow funds to the seller if the buyer takes no action for 3 days after the "Delivered" status is set.
  - Commission Deduction: The system must automatically calculate and route a 0.5% commission fee to the Admin wallet upon a successful transaction.

3.2.2 Non-Functional Requirements

These requirements define the quality attributes, performance constraints, and security measures of the system.

- **Security**
  - Data Protection: User passwords must be securely hashed (using Bcrypt) before storage in the database.
  - Authentication Tokens: The system must use secure JWT (JSON Web Tokens) to authorize API requests and validate WebSocket connections.
  - Role-Based Access Control: Backend API routes must enforce strict authorization checks to prevent unauthorized data manipulation (e.g., a buyer cannot access admin dispute routes).
- **Performance & Efficiency**
  - Real-Time Responsiveness: The WebSocket chat and notification system must deliver messages, offers, and UI cards instantly without requiring the user to refresh the browser page.
  - Concurrency: The database (SQLite/PostgreSQL) and SQLAlchemy ORM must safely handle multiple users modifying inventory and sending messages simultaneously without data corruption.

- **Usability**
  - Interface: The user interface must feature a modern, premium design with dynamic animations, distinct status badges, and intuitive navigation that requires zero user training.
  - Asynchronous Feedback: The frontend must provide immediate visual feedback (e.g., toast notifications, changing button states to "Declining...") to keep users informed during network requests.

- **Reliability & Availability**
  - Uptime & Stability: The backend server and WebSocket Connection Manager must maintain continuous uptime to ensure chat sessions do not arbitrarily disconnect.
  - Data Integrity: The database schema must enforce referential integrity (e.g., an offer cannot exist if the associated product or chat session is deleted).

3.3 Use Case Diagram

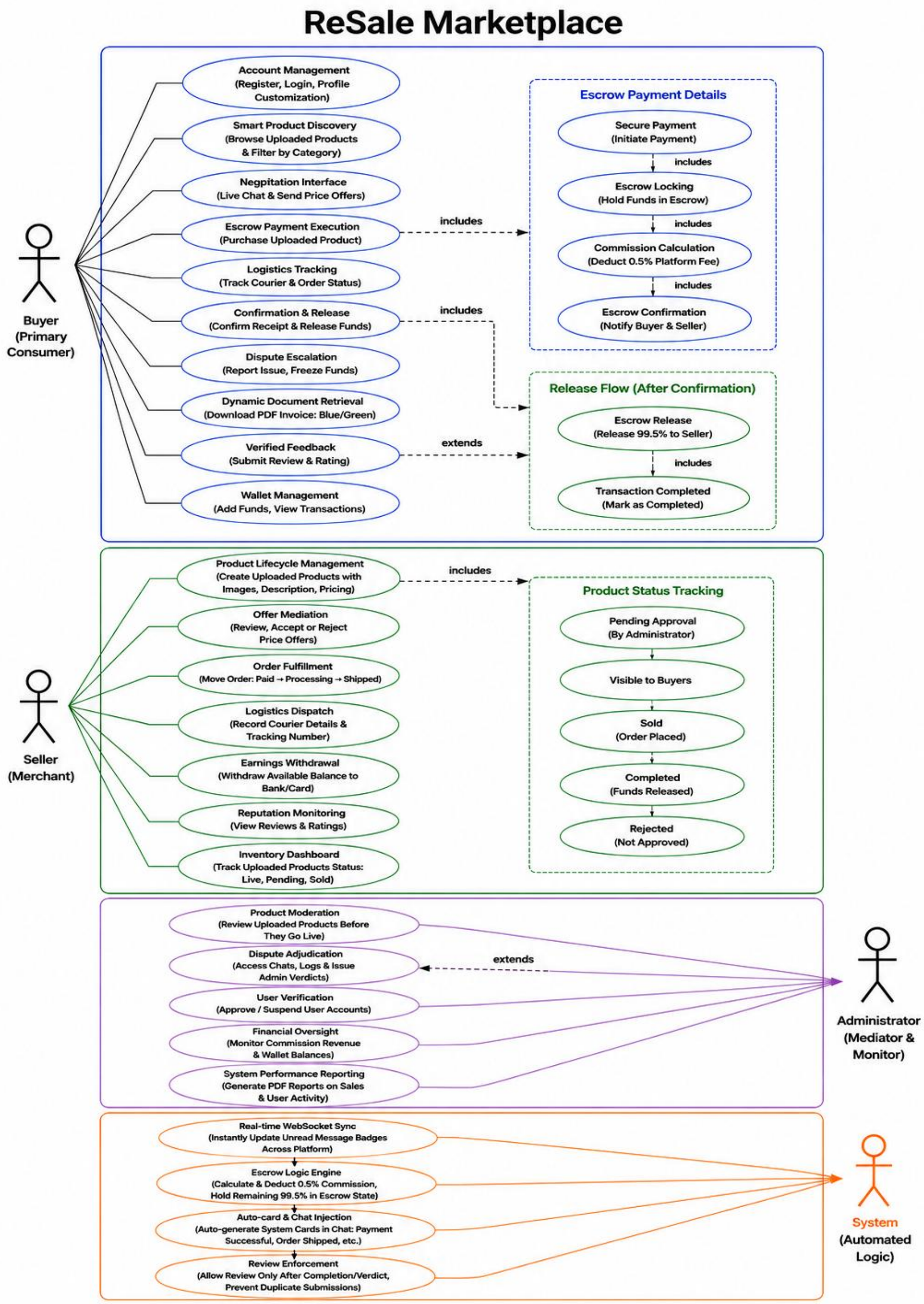


Figure 3.1 Use Case Diagram

Chapter 4.

Analysis

4.1 Activity Diagram

4.1.1 Buyer Operational Activity Diagram

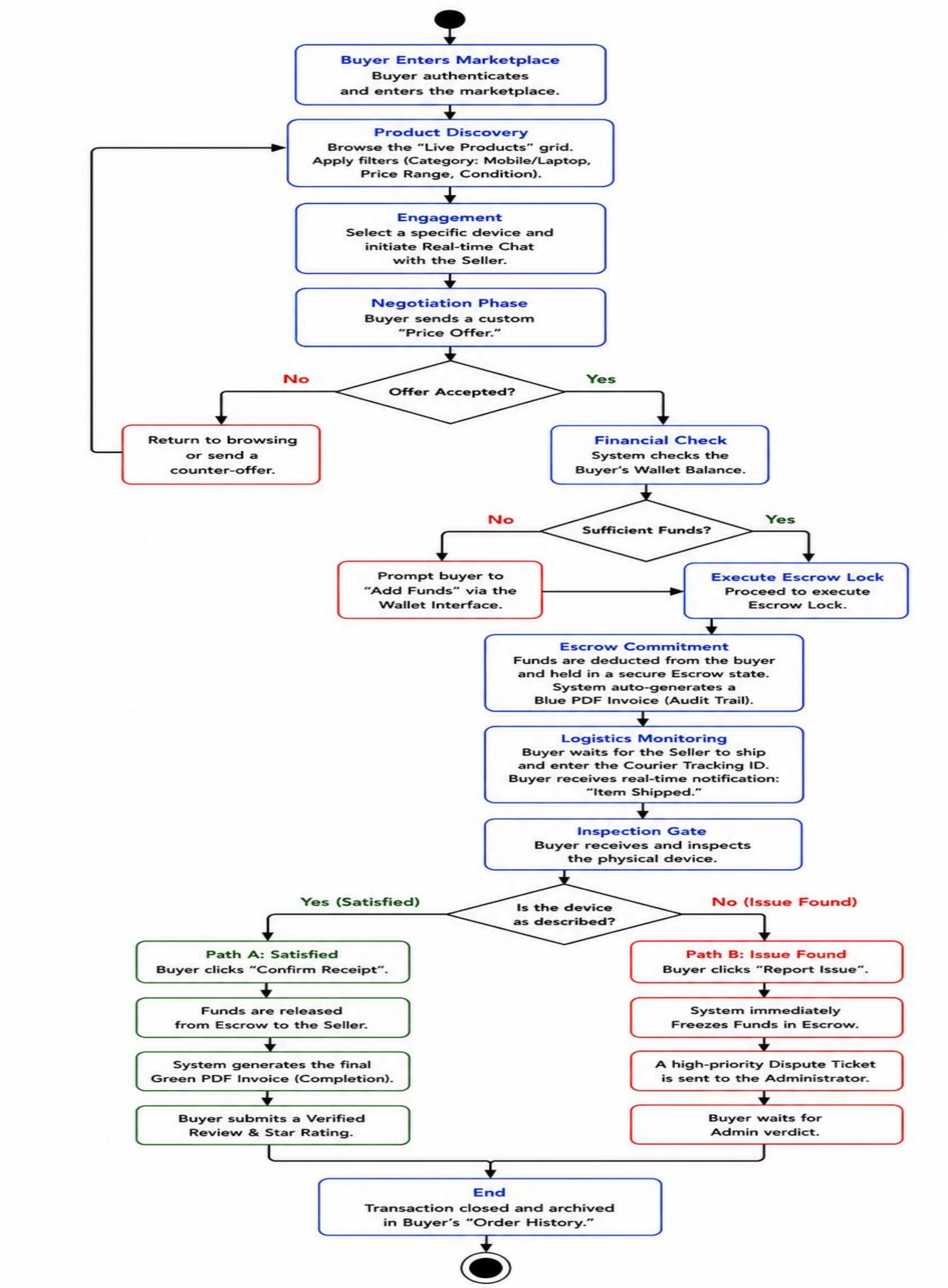


Figure 4.1 Buyer Operational Activity Diagram

4.1.2 Seller Operational Activity Diagram

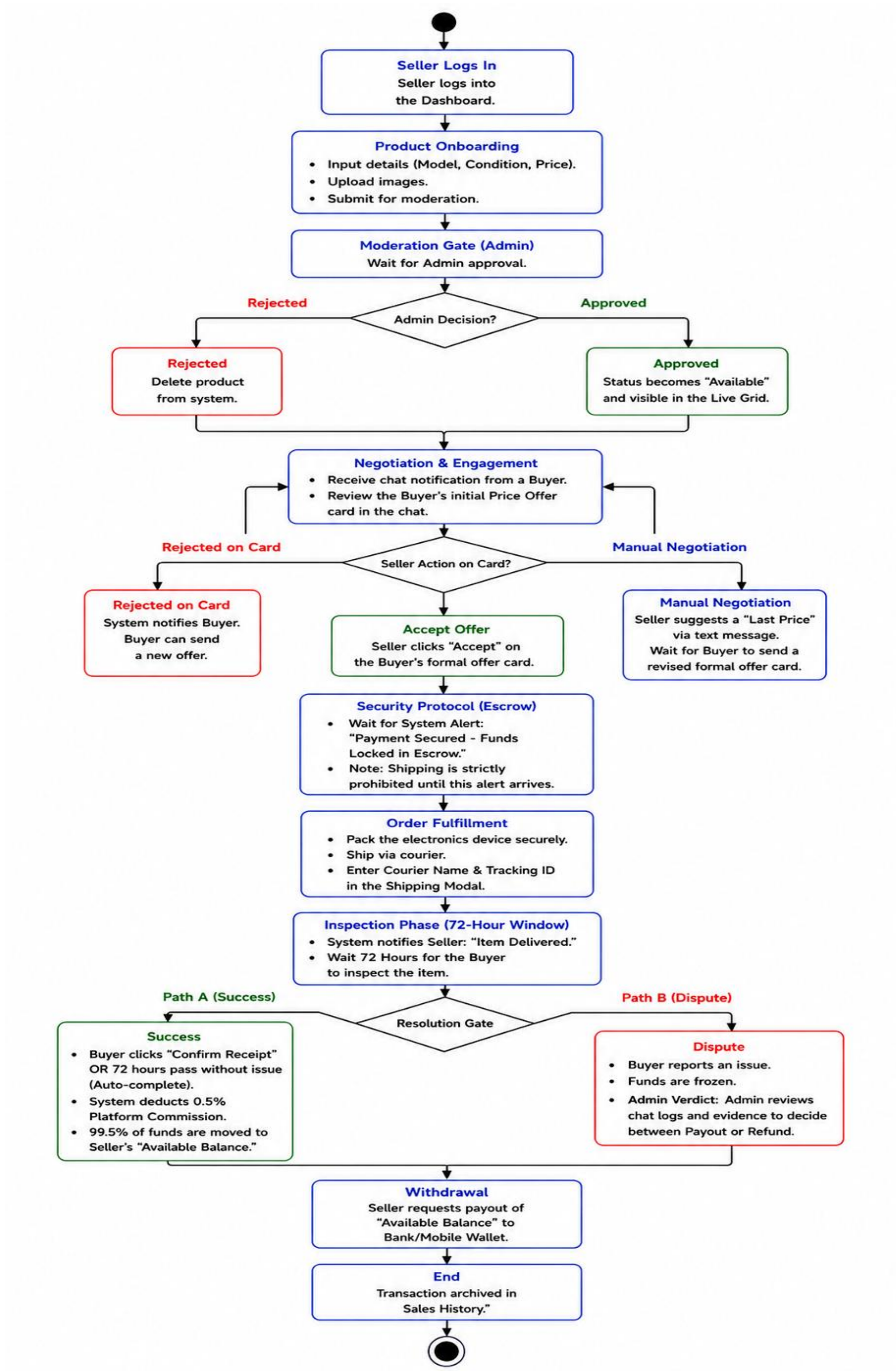


Figure 4.2 Seller Operational Activity Diagram

4.1.3 Admin Operational Activity Diagram

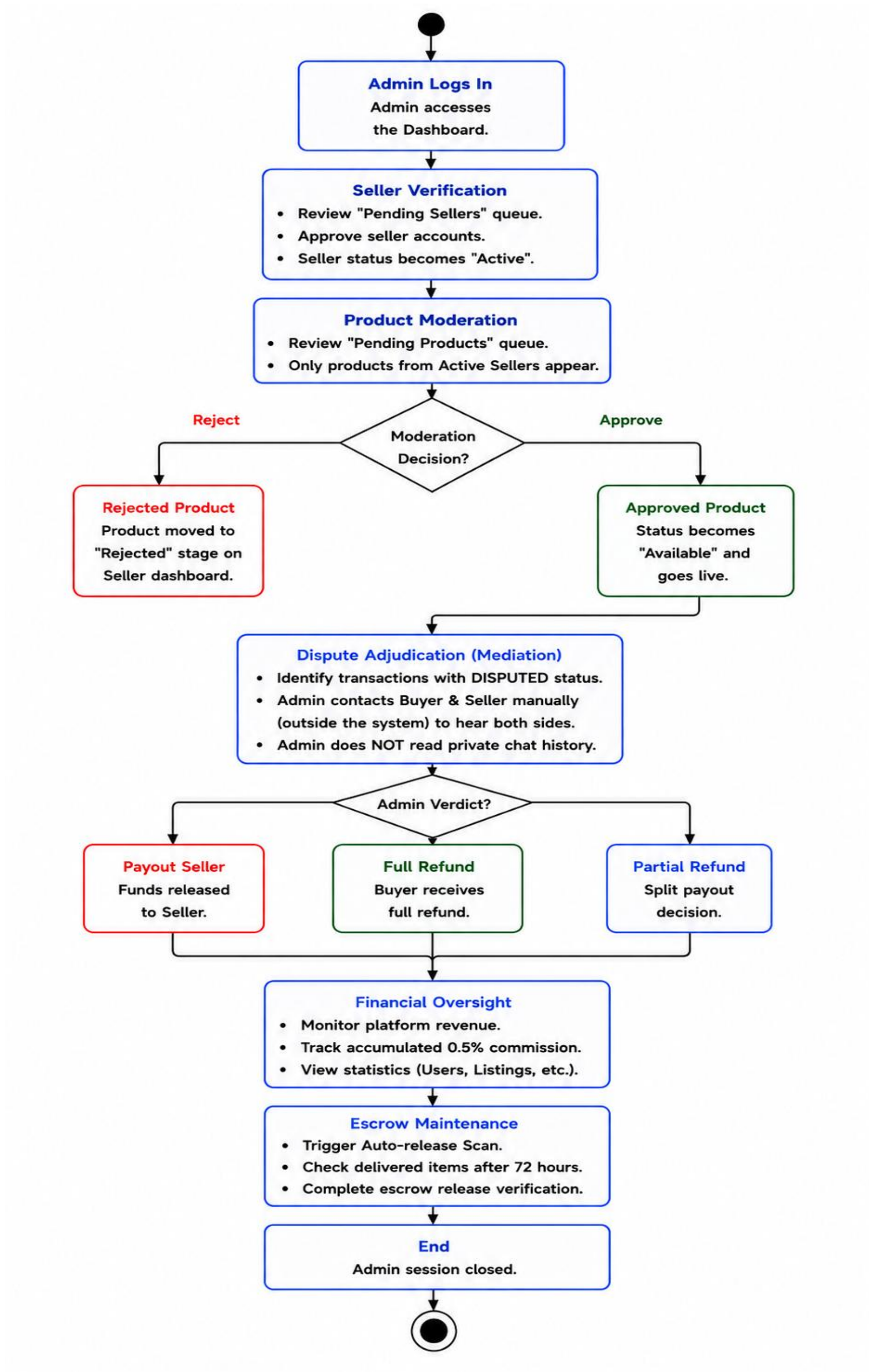


Figure 4.3 Admin Operational Activity Diagram

4.1.4 Admin Dispute Resolution Activity Diagram

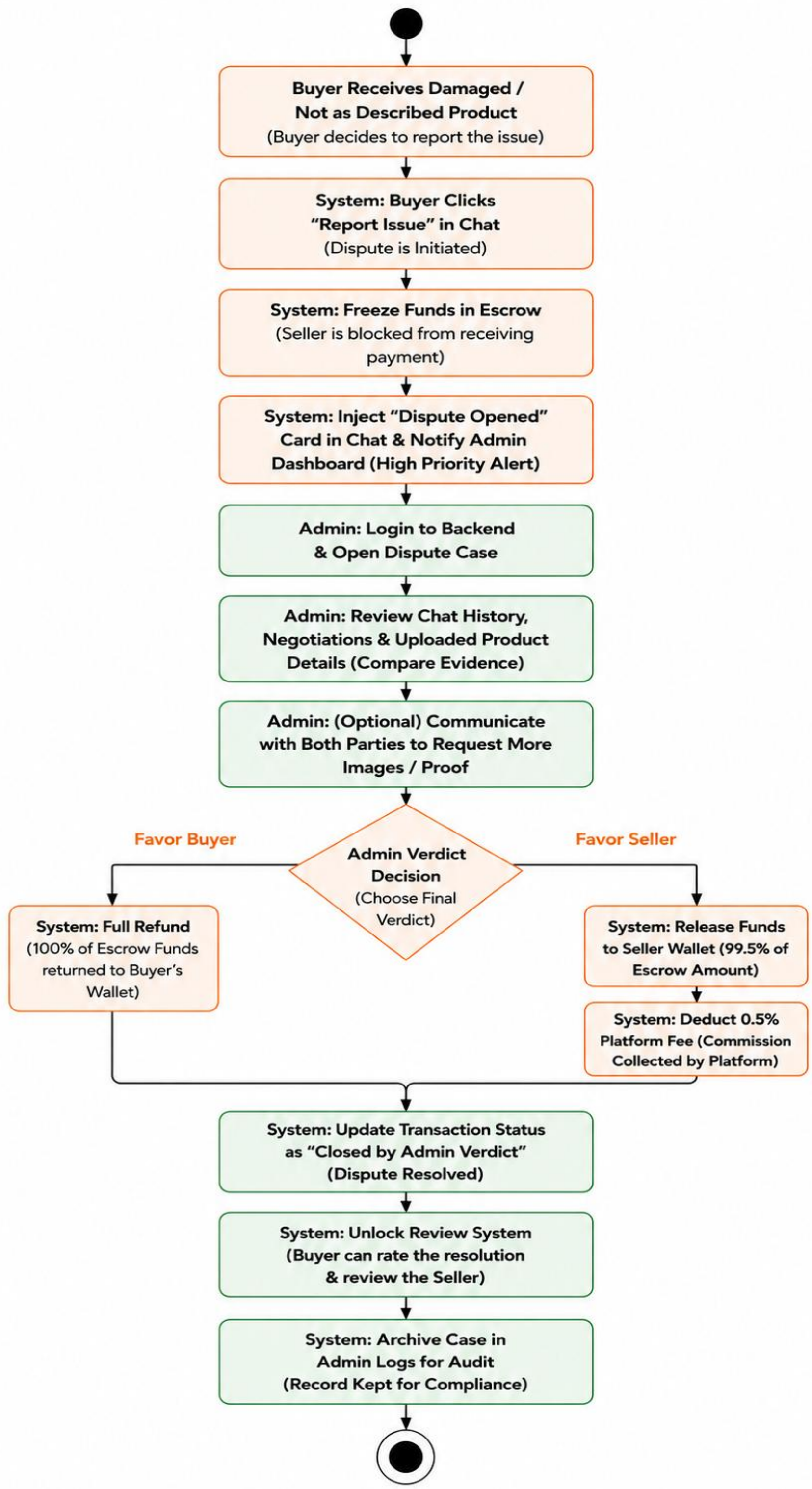


Figure 4.4 Admin Dispute Resolution Activity Diagram

4.1.5 Escrow Transaction Activity Diagram

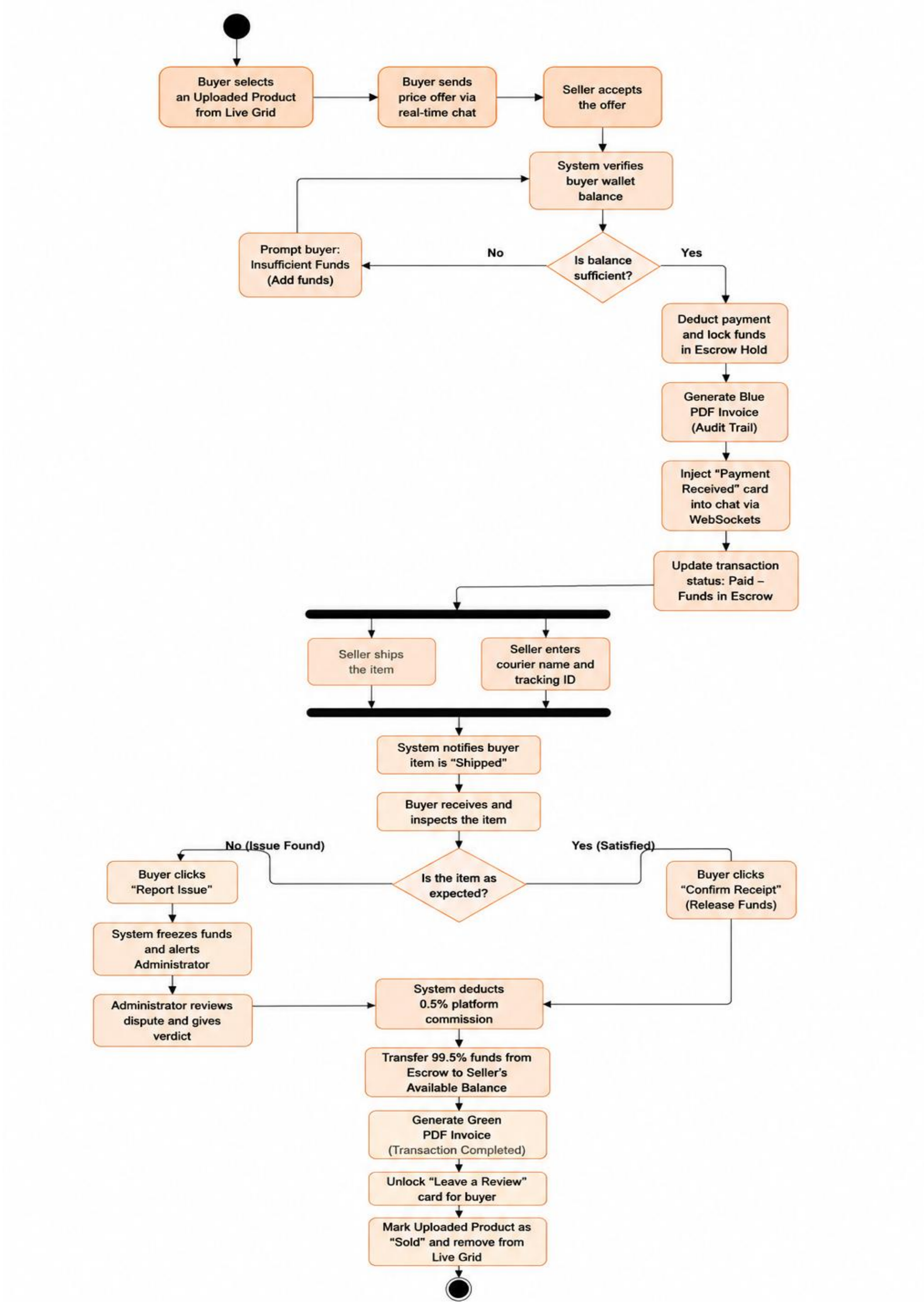


Figure 4.5 Escrow Transaction Activity Diagram

4.1.6 Product Life Cycle Activity Diagram

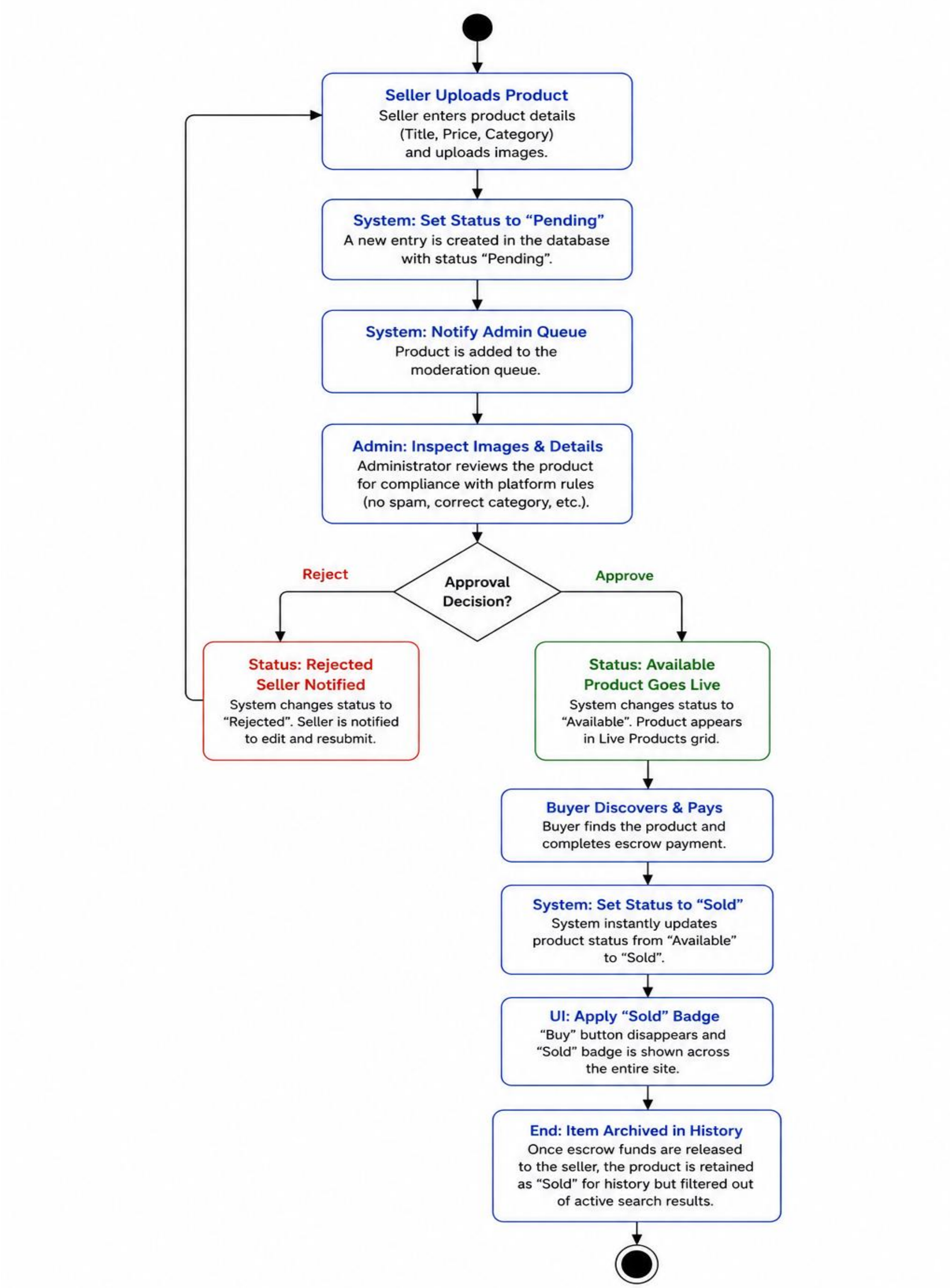


Figure 4.6 Product Life Cycle Activity Diagram

4.2 Swimlane Diagram

4.2.1 ReSale Marketplace Swimlane Diagram

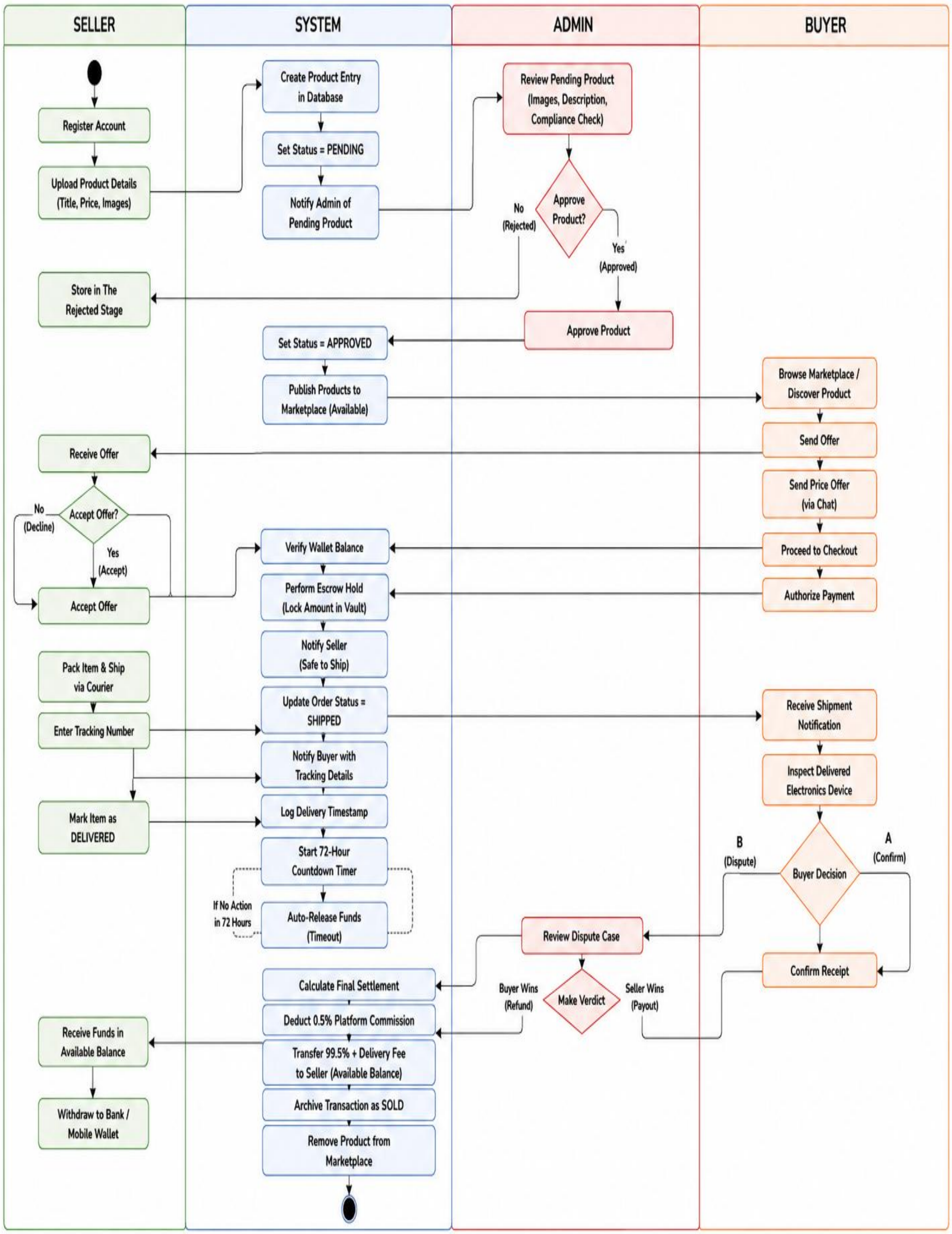


Figure 4.7 ReSale Marketplace Swimlane Diagram

4.2.2 Negotiation & Escrow Lock Swimlane Diagram

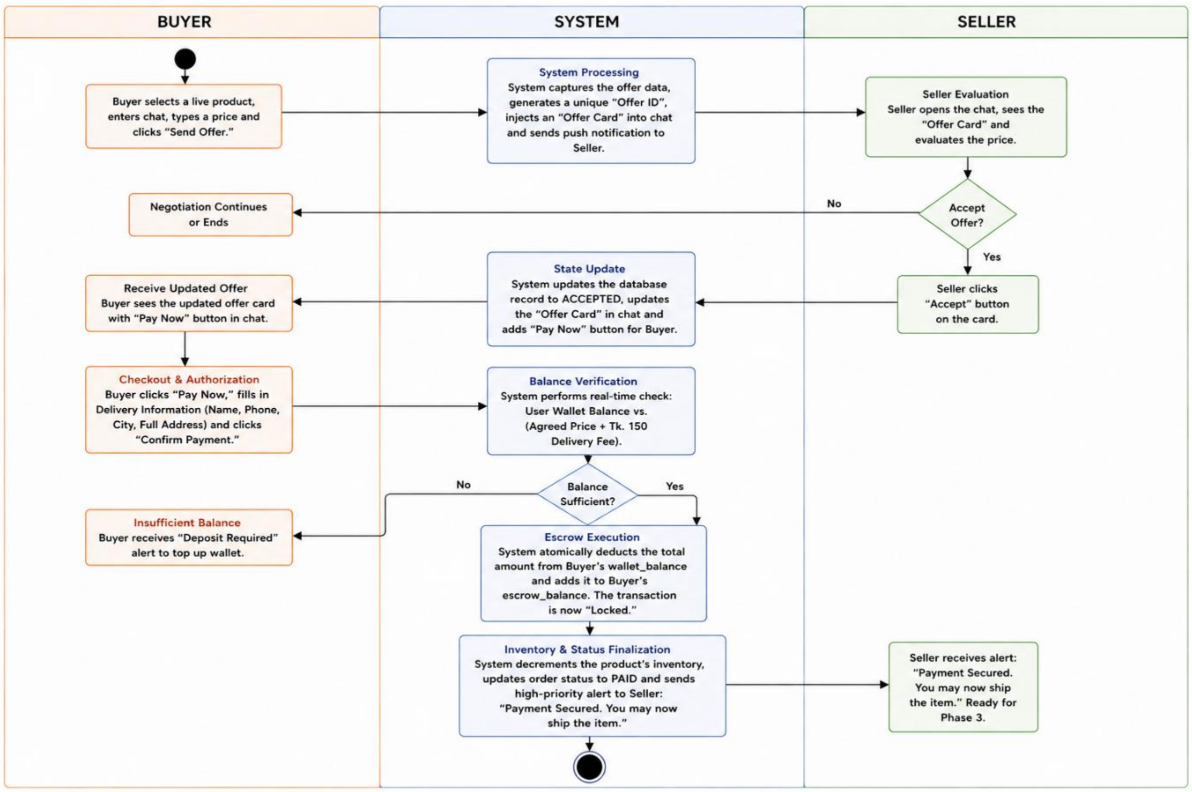


Figure 4.8 Negotiation & Escrow Lock Swimlane Diagram

4.2.3 Logistic & Order Processing Swimlane Diagram

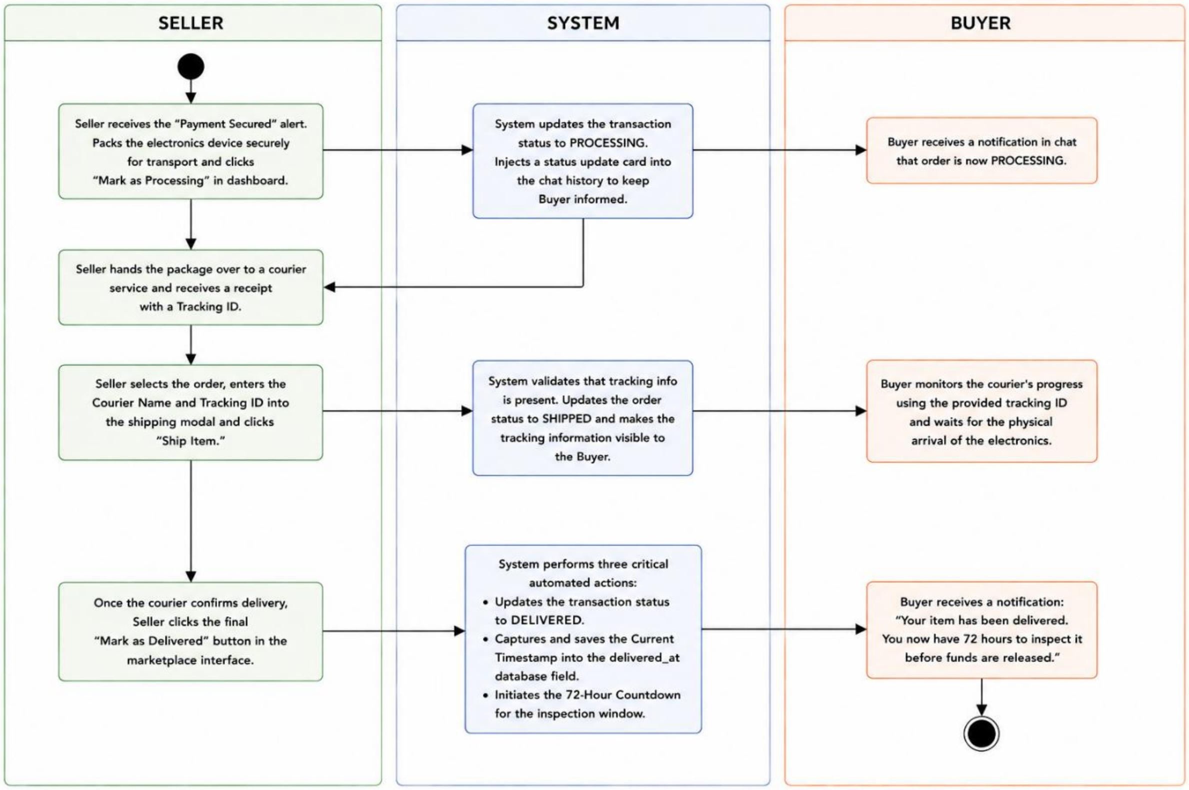


Figure 4.9 Logistic & Order Processing Swimlane Diagram

4.2.4 Resolution and Settlement Swimlane Diagram

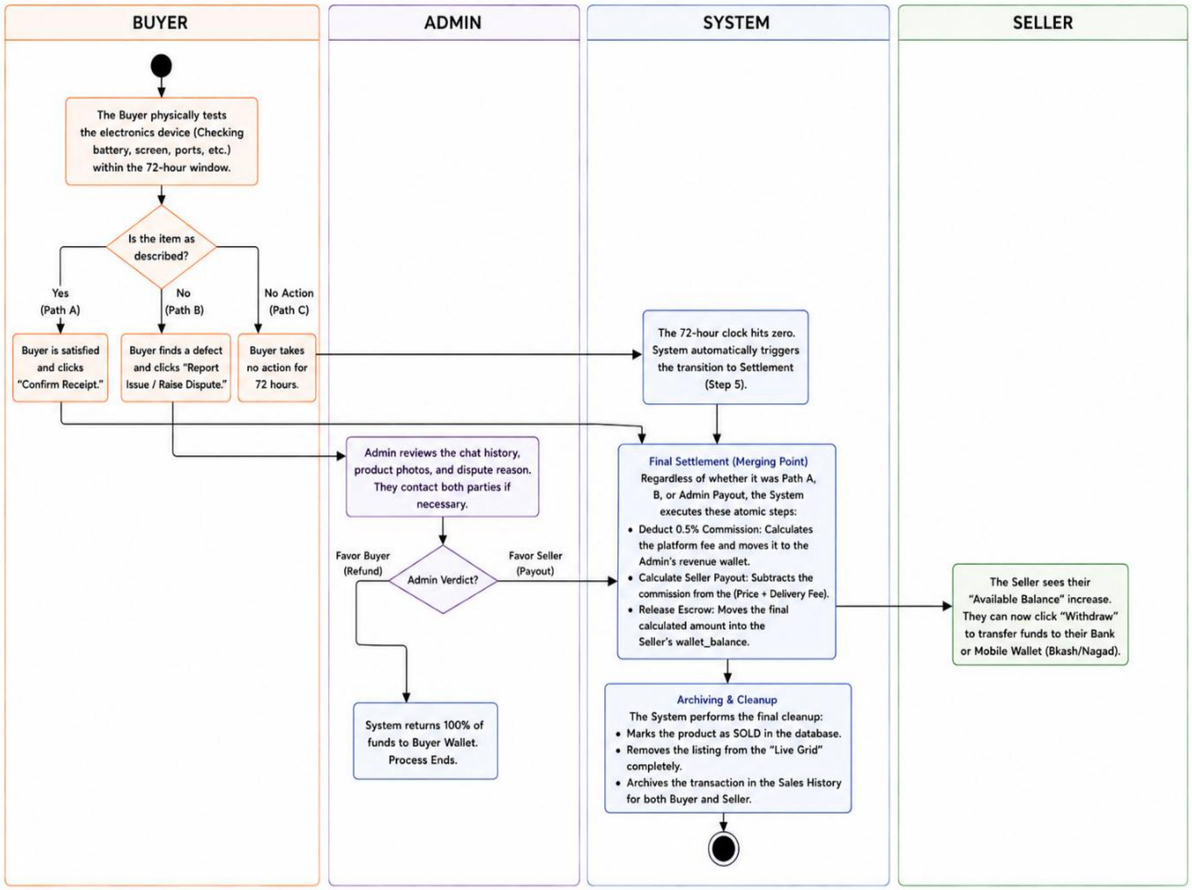


Figure 4.10 Resolution and Settlement Swimlane Diagram

4.3 Class Diagram

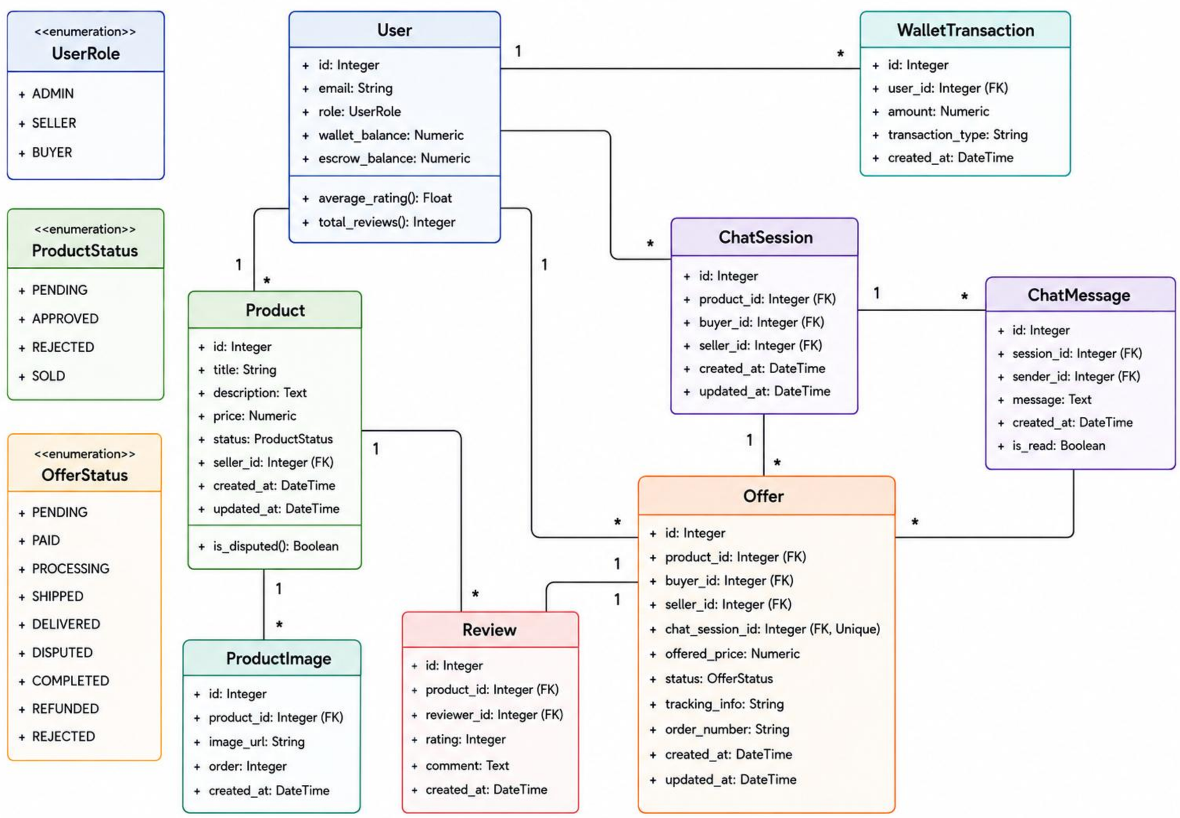


Figure 4.11 Class Diagram of Resale Marketplace

4.4 Sequence Diagram

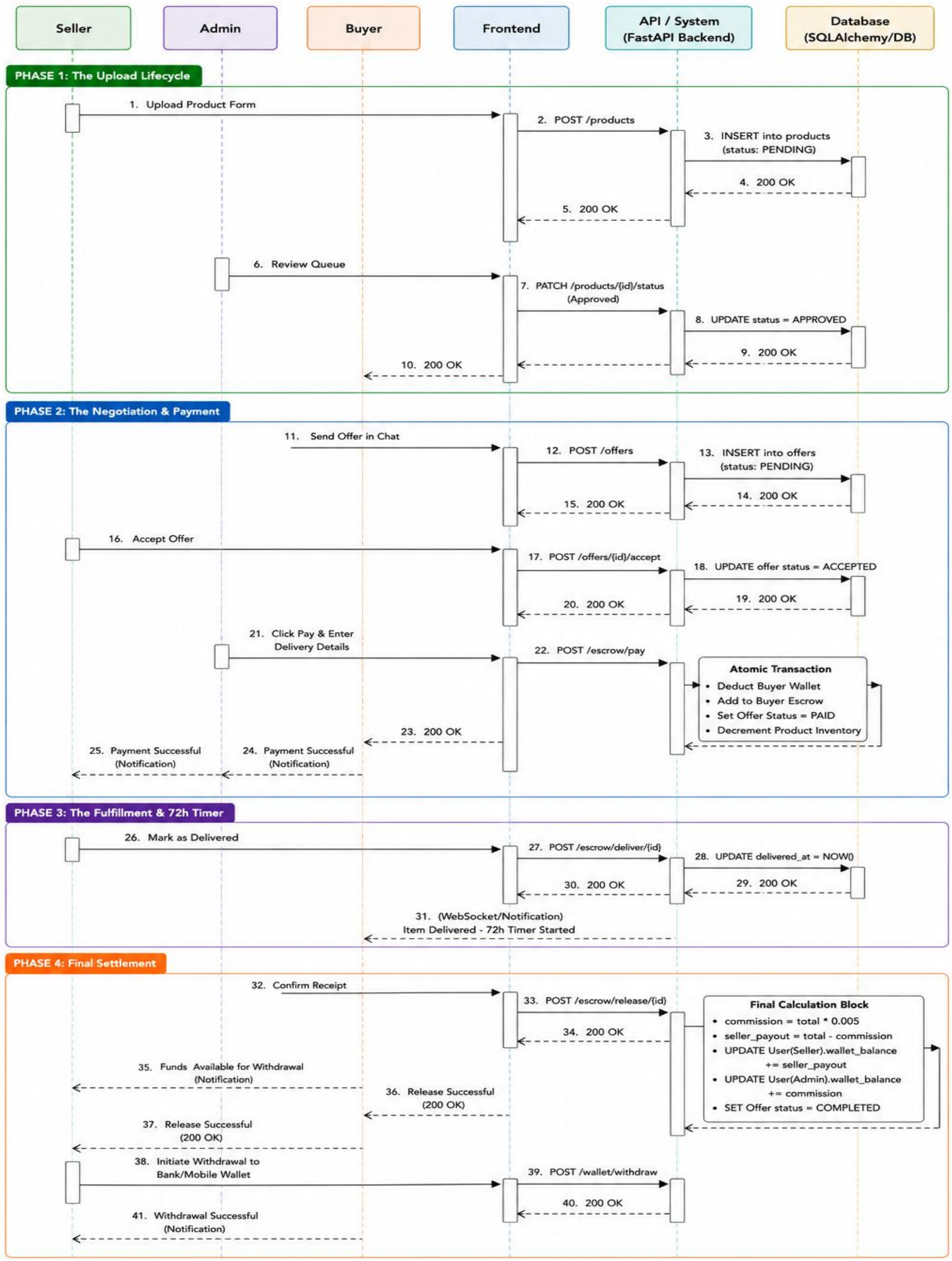


Figure 4.12 Sequence Diagram of ReSale Marketplace

4.5 CRC Cards (Class-Responsibility-Collaborator)

The CRC Cards offer fine-grained, object-oriented model of system components. Each card describes one class's Responsibilities and its Collaborators.

Table 4.1 CRC of User

CLASS: USER

RESPONSIBILITIES	COLLABORATORS
• Maintains user profile data (Email, Name, Phone, Address).	• **Product** Collaborates when the user acts as a seller to create and manage products.
• Manages secure authentication (Hashed Passwords).	• **Review** Provides reviews about the user; used to calculate average rating and total reviews.
• Tracks financial balances (wallet_balance and escrow_balance).	• **WalletTransaction** Records all financial transactions (deposits, withdrawals, fees, payouts) related to the user.
• Calculates user reputation (Average Rating and Total Reviews).	• **Offer** Linked when the user participates in offers as a buyer or seller.
• Maintains account status (Active, Banned, or Suspended).	
• Acts as Buyer or Seller in the marketplace.	
• Manages withdrawal and escrow operations.	

Table 4.2 CRC of Product Image

CLASS: PRODUCTIMAGE

RESPONSIBILITIES	COLLABORATORS
• Stores relative file paths to uploaded images on the server.	• **Product (The parent item)** The product to which this image belongs.
• Maintains the display order of photos.	
• Identifies the "Cover Photo" (Order = 0).	

Table 4.3 CRC of Product

CLASS: PRODUCT	
RESPONSIBILITIES	**COLLABORATORS**
• Stores electronics metadata (Model, Category, Description). • Manages pricing and physical condition details. • Tracks current listing status (Pending, Approved, Sold). • Manages available inventory quantity.	• **User (The Seller)** The seller who lists and manages this product. • **ProductImage (Multiple photos)** Stores multiple images associated with this product. • **Review (Feedback for this item)** Contains reviews and ratings provided by buyers. • **Offer (Purchase requests)** Contains offers or purchase requests made by buyers for this product.

Table 4.4 CRC of Chat Message

CLASS: CHATMESSAGE	
RESPONSIBILITIES	**COLLABORATORS**
• Stores individual text messages or system alerts. • Records the exact timestamp of communication. • Identifies the sender of each message. • Tracks if a message has been seen by the recipient.	• **ChatSession (The parent container)** The chat session that contains this message. • **User (The sender)** The user who sends this message.

Table 4.5 CRC of Chat Session

CLASS: CHATSESSION	
RESPONSIBILITIES	**COLLABORATORS**
• Creates a unique communication channel between a Buyer and Seller. • Links a specific Product to a negotiation thread. • Tracks the "Read/Unread" status of the conversation. • Manages "soft deletion" flags for both parties.	• **User (Participants)** The buyer and seller who participate in the conversation. • **Product (The subject of chat)** The product that is being discussed in this chat session. • **ChatMessage (The contents)** The messages exchanged within this chat session. • **Offer (Negotiated prices)** The offers created and shared during the negotiation in this chat.

Table 4.6 CRC of Offer

CLASS: OFFER	
RESPONSIBILITIES	**COLLABORATORS**
• Acts as the primary contract for a transaction. • Stores the final negotiated price and quantity. • Manages the order status (Paid, Shipped, Delivered, Disputed). • Stores critical tracking info and delivery addresses. • Triggers the automated Escrow fund release logic.	• **Product (The item being bought)** The product that is the subject of this offer. • **User (The specific Buyer and Seller)** The buyer who makes the offer and the seller who receives it. • **ChatSession (The origin of the offer)** The chat session where the offer was created. • **WalletTransaction (Recording the payment)** Records the payment transaction made for this offer.

Table 4.7 CRC of Review

CLASS: REVIEW	
RESPONSIBILITIES	**COLLABORATORS**
• Stores numerical ratings (1-5 stars) for a transaction. • Maintains written feedback/comments from the Buyer. • Links a specific transaction to a Seller's public reputation.	• **User (The Reviewer and the Recipient)** The user who writes the review (buyer) and the user who receives it (seller). • **Product (The item reviewed)** The product that was part of the transaction being reviewed.

Table 4.8 CRC of Wallet Transaction

CLASS: WALLETRANSACTION	
RESPONSIBILITIES	**COLLABORATORS**
• Acts as a permanent ledger for all financial movements. • Logs transaction types (Deposit, Withdrawal, Escrow Hold, Payout). • Stores exact amounts and timestamps for auditing. • Maintains a description of the event (e.g., "Payment for iPhone 13").	• **User (The owner of the wallet)** The user who owns the wallet and is associated with this transaction.

Chapter 5

Risk Management

5.1 Risk Management

Risk management is the use of available resources to undertake the activities of risk identification, risk analysis, and risk response planning. In an electronics market, trust and financial security matter the most. The plan allows the group to expect risks involved in escrow breaches, product scams, and servers crashing for a safe and productive trade environment.

5.1.1 Stages of Risk Management

a. **Risk Identification**: Finding potential threats to the marketplace.

b. **Risk Analysis**: Evaluating the likelihood and impact of each threat.

c. **Planning**: Developing strategies to mitigate or avoid risks.

d. **Monitoring**: Continuous oversight to catch risks before they become issues.

5.2 Risk Identification

The following table identifies the major risks associated with the Resale Marketplace:

Table 5.1 Risk Identification

Risk ID	Risk Type	Possible Risk
R1	Technical	System crashes during negotiation, WebSocket disconnection, Database locking.
R2	Security	Unauthorized access to Admin panel, Account takeover, JWT token theft.
R3	Financial	Escrow fund lock failure, payment gateway errors, commission calculation bugs.
R4	Operational	Product Fraud (Counterfeit electronics), "Item not as described" disputes.

R5	Legal/Compliance	Consumer protection laws, data privacy (GDPR/Local laws), tax compliance.
R6	Project Management	Scope creep, delays in implementing the 72-hour auto-release logic.
R7	User Experience	Complex dispute filing process, slow loading of high-res product images.
R8	Integration	Payment API (Secure Pay) failures, SMS/Email notification delays.
R9	Data Management	Loss of transaction audit trails, image storage corruption.
R10	Performance	High concurrency during peak buying hours, real-time chat latency.

5.3 Risk Analysis

The risks identified for the ReSale Marketplace are analyzed based on their probability of occurrence. This analysis helps prioritize risk mitigation and management strategies during system development and operation.

Table 5.2 Risk Analysis

Possible Risks	Probability	Impact
Escrow/Financial Failure	15-25% (Low)	Catastrophic
Product Fraud (Seller Scams)	35-45% (High)	Serious
Data Admin Hacking	20-30% (Moderate)	Catastrophic

System Crashes/Downtime	25-35% (Moderate)	Serious
Dispute Resolution Delays	30-40% (Moderate)	Tolerable
Project Delivery Delays	30-40% (Moderate)	Serious
Payment Gateway Failures	15-20% (Low)	Serious
Integration/API Failures	20-25% (Moderate)	Serious
Scalability/Concurrency	30-40% (Moderate)	Serious
Data/Audit Loss	10-15% (Low)	Serious

5.4 Risk Planning

Risk planning defines the strategies and action plans used to handle risks if they occur during system operation. The following table demonstrate the risk planning of ReSale Marketplace:

Table 5.3 Risk Planning

Risk ID	Risk	Strategy
R1	Financial / Escrow Failure	Use ACID-compliant database transactions for all fund movements. Implement dual-entry logging for wallet and escrow balances.
R2	Security Breach / Admin Hacking	Implement Role-Based Access Control (RBAC), 2FA for Admins, JWT encryption, and regular security audits of the API.
R3	Product Fraud / Seller Scams	Mandatory Admin moderation of all electronics listings, Seller verification system, and the 72-hour funds-hold period.

R4	System Downtime	Cloud hosting with automated scaling, load balancing, and real-time health monitoring of the FastAPI backend.
R5	Legal / Compliance Issues	Clear Terms of Service, Privacy Policy for user data, and a transparent Escrow policy to meet consumer protection standards.
R6	Project Delivery Delay	Agile Sprints, prioritized backlog (MVP focus), and daily standups to identify implementation blockers early.
R7	Poor User Experience	Intuitive Chat-UI for negotiation, mobile-responsive design, and a simplified "One-Click" dispute filing system.
R8	Third-party API Failures	Implement retry logic with exponential back off for Payment/SMS APIs and fallback notification channels.
R9	Performance / Scalability	Caching for product grids, optimized SQL queries for transactions, and CDN for serving high-resolution electronics images.

5.5 RMMM (Risk Mitigation Management and Monitoring) Planning

Throughout the development lifecycle, risks are methodically tracked. The following tables detail the RMMM planning for the highest priority risks in the Resale Marketplace.

Table 5.4 RMMM Planning for Risk 1 (Financial Security)

Project Risk 1	Details
Risk Name	Escrow failure or fund calculation errors.
Probability	15-25% (Low)
Impact	Catastrophic
Description	Errors in moving funds from Buyer wallet to Escrow or incorrect 0.5% commission calculation.
Mitigation & Monitoring	Use database transactions (commit/rollback), automated unit tests for financial logic, and real-time transaction logging.
Management	Finance/Tech team freezes affected accounts and manually audits the ledger to correct discrepancies.
Status	Managed

Table 5.5 RMMM Planning for Risk 2 (Marketplace Fraud)

Project Risk 2	Details
Risk Name	Product Fraud (Counterfeit items or empty boxes).
Probability	35-45% (High)
Impact	Serious
Description	Sellers shipping items that don't match descriptions or committing intentional scams.

Mitigation & Monitoring	72-hour escrow hold, mandatory listing photos, and Admin moderation of all high-value electronics.
Management	Admin reviews dispute evidence (photos/chat logs) and executes a full refund to the Buyer if fraud is proven.
Status	Managed

Table 5.6 RMMM Planning for Risk 3 (Unauthorized Access)

Project Risk 3	Details
Risk Name	Data breaches or Admin dashboard compromise.
Probability	20% – 30% (Moderate)
Impact	Catastrophic
Description	Unauthorized access to user sensitive data or the ability to manipulate escrow funds via Admin panel.
Mitigation & Monitoring	SSL/TLS encryption, strong RBAC (Role-Based Access Control), and periodic token rotation.
Management	Immediate revocation of compromised tokens, system lockdown, and forensic log analysis.
Status	Managed

Table 5.7 RMMM Planning for Risk 4 (Real-time Latency)

Project Risk 4	Details
Risk Name	WebSocket or Chat disconnection.
Probability	30% – 40% (Moderate)
Impact	Tolerable
Description	Buyers and Sellers unable to negotiate in real-time due to connection drops.
Mitigation & Monitoring	Automatic WebSocket reconnection logic on the frontend and "Last Seen" message indicators.
Management	Support team advises users to refresh or uses persistent HTTP polling as a fallback.
Status	Managed

Table 5.8 RMMM Planning for Risk 5 (Legal Compliance)

Project Risk 5	Details
Risk Name	Non-compliance with consumer protection or data laws.
Probability	15% – 25% (Low)
Impact	Serious
Description	Legal penalties due to improper handling of customer funds or data privacy violations.

Mitigation & Monitoring	Periodic legal review of the Escrow workflow and transparent user data management.
Management	Legal team updates Terms of Service and adjusts system logic to comply with new regulations.
Status	Managed

Table 5.9 RMMM Planning for Risk 5 (Dispute Bottlenecks)

Project Risk 6	Details
Risk Name	Item Not as Described" or Dispute Bottlenecks
Probability	30-40% (Moderate)
Impact	Tolerable
Description	High volume of disputes exceeding Admin capacity, leading to slow fund releases and user frustration.
Mitigation & Monitoring	Standardized dispute forms requiring photo evidence. Dashboard alerts for any dispute exceeding 48 hours without an admin response
Management	Temporarily increase admin staff or prioritize disputes based on the financial value of the electronics involved.
Status	Managed

Table 5.10 RMMM Planning for Risk 5 (Scope Creep)

Project Risk 7	Details
Risk Name	Scope Creep or Technical Implementation Delays
Probability	30-40% (Moderate)
Impact	Serious
Description	Failure to meet deadlines for complex features like the 72-hour auto-release logic or invoice generation.
Mitigation & Monitoring	Use Agile Sprints with a strict focus on the Minimum Viable Product (MVP). Daily standups to identify and clear technical "blockers" early
Management	Pivot resources to core transaction features and postpone non-essential UI polish to secondary release phases.
Status	Managed

Table 5.11 RMMM Planning for Risk 5 (Database Locking)

Project Risk 8	Details
Risk Name	Real-time Latency or Database Locking
Probability	30-40% (Moderate)
Impact	Serious
Description	High concurrency during peak hours causing slow database queries or lag in the real-time chat negotiation system.

Mitigation & Monitoring	Optimize SQL indexes for transaction tables and implement Redis caching for the product feed. Monitor query execution times.
Management	Vertical scaling of database instances and optimizing long-running queries identified by the performance logs.
Status	Managed

Table 5.12 RMMM Planning for Risk 5 (Data Corruption)

Project Risk 9	Details
Risk Name	Loss of Transaction Audit Trails or Data Corruption
Probability	10-15% (Low)
Impact	Serious
Description	Failure of database storage or corruption of transaction history, making it impossible to verify old escrow releases or disputes.
Mitigation & Monitoring	Daily automated database backups stored in a secondary cloud location. Implementation of Write-Ahead Logging (WAL) and checksums for critical transaction tables.
Management	Restore the system from the most recent 24-hour backup and re-sync any missing transactions using the separate log of processed payment gateway callbacks.
Status	Managed

Table 5.13 RMMM Planning for Risk 5 (Data Privacy Violations)

Project Risk 9	Details
Risk Name	Data Privacy Violations (GDPR/Local Laws)
Probability	10-15% (Low)
Impact	Serious
Description	Improper handling of user personal data (PII) leading to legal penalties or loss of user trust.
Mitigation & Monitoring	Encrypt sensitive data at rest (passwords/phone numbers). Regular privacy audits and ensuring the "Right to be Forgotten" (account deletion) is fully implemented.
Management	Legal counsel review of the data leak/violation. Transparent communication with affected users and system updates to patch the vulnerability.
Status	Managed

Chapter 6.

Project Planning and Scheduling

6.1 Project planning and scheduling

Planning and Scheduling for Development of the Resale Marketplace for Electronics Devices Planning and Scheduling are crucial for the success in developing the Resale Market place for Electronic Devices. Project Planning Project planning is the activity of defining the scope and objectives of a project, identifying the work needed to execute the project, and analyzing the risk. This is what I mean when I say our engineering team has a clear roadmap for developing a safe, efficient market for trading electronics. Putting all these activities together with a time-base and scheduling to include milestones on real-time chat, escrow payments and product verification Project scheduling, however, sets all these activities into a time framework and creates Milestones for certain features, such as pausing all product verification, escrow payments, etc. They can be combined to allow a disciplined, high-quality delivery of the platform within the agreed timescale Planning and scheduling play a crucial role in brokering the Resale Marketplace for Electronics Devices. Project planning is the act of defining the scope and objectives of a project, identifying the resources, the necessary work, and risks. This means our engineering team has a solid plan to work from in creating a safe, effective market for buying and selling electronics. Now, project scheduling takes these activities and sets them in a time-based sequence, establishing milestones for real-time chat, escrow settlements, and product verification. As a duo, they are able to facilitate a focused high-level execution of the platform in a timely fashion.

6.2 Function Point Estimation

Function Point Analysis (FPA) is utilized to measure the functional size of the Resale Marketplace. By calculating function points, we estimate the project's complexity, the effort required for development, and the overall cost.

6.2.1. Functionality, Input, Output

Table 6.1 Functionality, Input and Output

Functionality	Input	Output
User Registration	Full Name, Email, Password, Phone, Address, Profile Pic, Role	Account created, confirmation email, redirect to login
User Authentication	Email, Password	JWT Access Token, Login success/failure message
Product Upload	Title, Category, Price, Condition, Description, Images, Inventory	Product pending approval, listing visible to sellers/admin
Product Search	Search Query, Category, Price Range, Condition	List of matching electronics products with details
Negotiation (Offers)	Offer Price, Quantity, Product ID, Session ID	Offer sent to seller, system message in chat
Real-time Chat	Message Text, Session ID, User ID	Message delivered to recipient, notification badge update
Payment & Escrow	Offer ID, Delivery Info, Payment Method	Funds held in escrow, Order ID generated, Seller notified
Wallet Operation	Deposit Amount, Withdrawal Method, Account Details	Wallet balance updated, transaction history log
Order Management	Offer ID, Status (Processing/Shipped/Delivered)	Order status updated, buyer notified, timeline updated
Escrow Release	Offer ID (Buyer Action)	Funds transferred to seller wallet, delivery fee processed

Review & Rating	Rating (1-5), Comment, Product ID	Seller rating updated, review visible on product page
Admin Dashboard	System Statistics, User Logs, Approval Requests	Analytics reports, User/Product management controls
Dispute Resolution	Dispute Reason, Evidence, Offer ID	Admin notified, payment hold extended, resolution status
Invoice Generation	Offer ID, Transaction Details	PDF Invoice with buyer/seller info and marketplace branding

6.2.2. Identify Complexity (Data Functions)

Data functions are categorized as Internal Logical Files (ILF) and External Interface Files (EIF).

Table 6.2 Identify Complexity of Data Function

Function Name	Types	Fields (DET)	RET	Complexity	UFP
Users	ILF	12	1	Low	7
Products	ILF	12	1	Low	7
Chat Sessions	ILF	7	1	Low	7
Chat Messages	ILF	6	1	Low	7
Product Images	ILF	4	1	Low	7
Reviews	ILF	7	1	Low	7
Wallet Transactions	ILF	6	1	Low	7

Offers (Orders)	ILF	18	1	Low	5
Payment Gateway (Bkash)	EIF	6	1	Low	5
Payment Gateway (Nagad)	EIF	6	1	Low	5
Email Service API	EIF	6	1	Low	5

6.2.3 Unadjusted Function Point Contribution

Transaction functions include External Inputs (EI), External Outputs (EO), and External Inquiries (EQ).

Table 6.3 Identify Complexity of Transaction function

Function Name	Type	Fields (DET)	FTR	Complexity	UFP
User Signup	EI	9	1	Low	3
User Login	EI	2	1	Low	3
Product Creation	EI	10	2	Average	4
Make Offer	EI	5	2	Average	4
Send Message	EI	3	2	Low	3
Escrow Payment	EI	8	3	High	6
Deposit Funds	EI	4	1	Low	3
Post Review	EI	5	2	Average	4
Admin Action	EI	4	2	Low	3

Buyer Dashboard	EO	15	4	High	7
Seller Dashboard	EO	15	4	High	7
Admin Dashboard	EO	25	8	High	7
Invoice PDF	EO	12	3	Average	5
Wallet Statement	EO	8	2	Average	5
Product Search	EQ	6	1	Low	3
View Order Status	EQ	5	2	Low	3
View Chat History	EQ	6	2	Average	4
Check Product Details	EQ	10	1	Low	3

6.2.3 Unadjusted Function Point Contribution

Table 6.4 Total Unadjusted Function Point (UFP)

Function Type	Count	Total UFP
External Input (EI)	9	33
External Output (EO)	5	31
External Inquiry (EQ)	4	13
Internal Logical File (ILF)	8	56
External Interface File (EIF)	4	20
Total Unadjusted Function Point (UFP)	30	153

6.2.4 Calculation of Total Degree of Influence (TDI)

Table 6.5 Calculation of Total Degree of Influence (TDI)

SI	General System Characteristics	Brief Description	DI
01	Data Communication	Requires real-time WebSocket communication for chat and notifications.	5
02	Distributed Data Processing	Hosted on cloud servers with image storage and database management.	3
03	Performance	Critical for real-time negotiations and secure financial transactions.	4
04	Heavily Used Configuration	High concurrent access for buyers and sellers during peak hours	4
05	Transaction Rate	Moderate to high due to active chat messaging and offer updates	4
06	Online Data Entry	All interactions (listing, chatting, and paying) are done via web interface.	5
07	End-User Efficiency	Focus on intuitive navigation for non-technical users to sell electronics.	5
08	Online Update	Real-time updates for wallet balances, order statuses, and messages.	5
09	Complex Processing	Escrow logic, fee calculation, and PDF invoice generation.	4

10	Reusability	Modular components for auth, chat, and wallet systems.	3
11	Installation Ease	Standard deployment on cloud platforms with environment configurations.	3
12	Operational Ease	Admin panels simplify product approval and dispute management.	4
13	Multiple Sites	Single platform accessible globally via web.	2
14	Facilitate Change	Scalable architecture to support more product categories and vendors.	4
	Total TDI		55

6.2.6. Final Calculation

- **Value Adjustment Factor (VAF)**

  VAF = 0.65 + (0.01 * TDI)

  VAF = 0.65 + (0.01 * 55)

  VAF = 1.20

- **Adjusted Function Point (AFP)**

  AFP = UFP * VAF

  AFP = 153 * 1.20

  AFP = 183.6

6.3 Project Scheduling

The development of the Resale Marketplace for Electronics Devices is structured into six distinct phases over a 12-week timeline. This staged methodology allows for iterative development, continuous quality assurance, and clear milestone tracking.

6.3.1 Development Phases

- Weeks 1-3 (Phase 1: Initiation & Planning): Focuses on gathering functional requirements and setting up the project architecture using FastAPI (backend) and Vanilla JavaScript (frontend).
- Weeks 3-5 (Phase 2: Design & Setup): Involves designing the relational database schema and creating UI/UX prototypes to ensure a premium user experience.
- Weeks 5-8 (Phase 3: Core Development): The primary development phase, covering User Authentication, Product Management (listing/search), and the Real-time Negotiation & Chat system.
- Weeks 8-10 (Phase 4: Advanced Features): Implementation of the Platform Wallet, Escrow Payment logic, PDF Invoice generation, and real-time Notification services.
- Weeks 10-11 (Phase 5: Testing & QA): Comprehensive system testing, including security audits of financial modules, bug fixes, and User Acceptance Testing (UAT).
- Weeks 11-12 (Phase 6: Deployment): Final deployment to production environments and user documentation/training.

6.3.2 Project Gantt Chart

The chart below visualizes the 12-week development roadmap and the overlapping nature of the design and development phases.

Figure 6.1 Resale Marketplace Development Roadmap

Chapter 7.

Designing

7.1 DFD (Data Flow Diagram)

A Data Flow Diagram (DFD) is a graphical tool used in software engineering to represent how data moves through a system, showing the flow between processes, data stores, and external entities. It helps us to easily identify how system is interchanging its data with database, system and stakeholders.

7.1.1 DFD Level 0 – Context Diagram of the Resale Marketplace

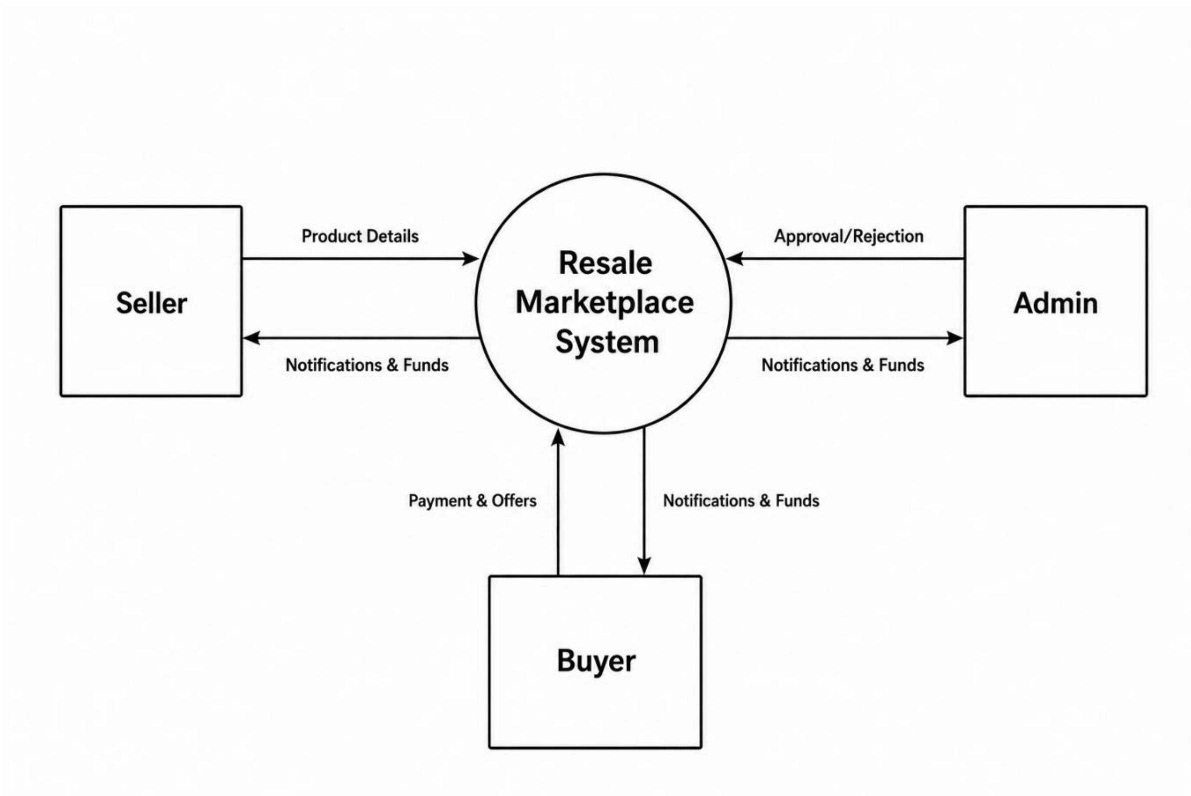


Figure 7.1 DFD Level 0 – Context Diagram

The Resale Marketplace Context Diagram (Level 0) describes the scope and the boundary of the Resale Marketplace. The Level 0 DFD represents a whole system as a single process and shows the interaction between that system and external entities such as the Buyer, the Seller, and the Admin.

7.1.2 DFD Level 1: Functional Decomposition of System Processes

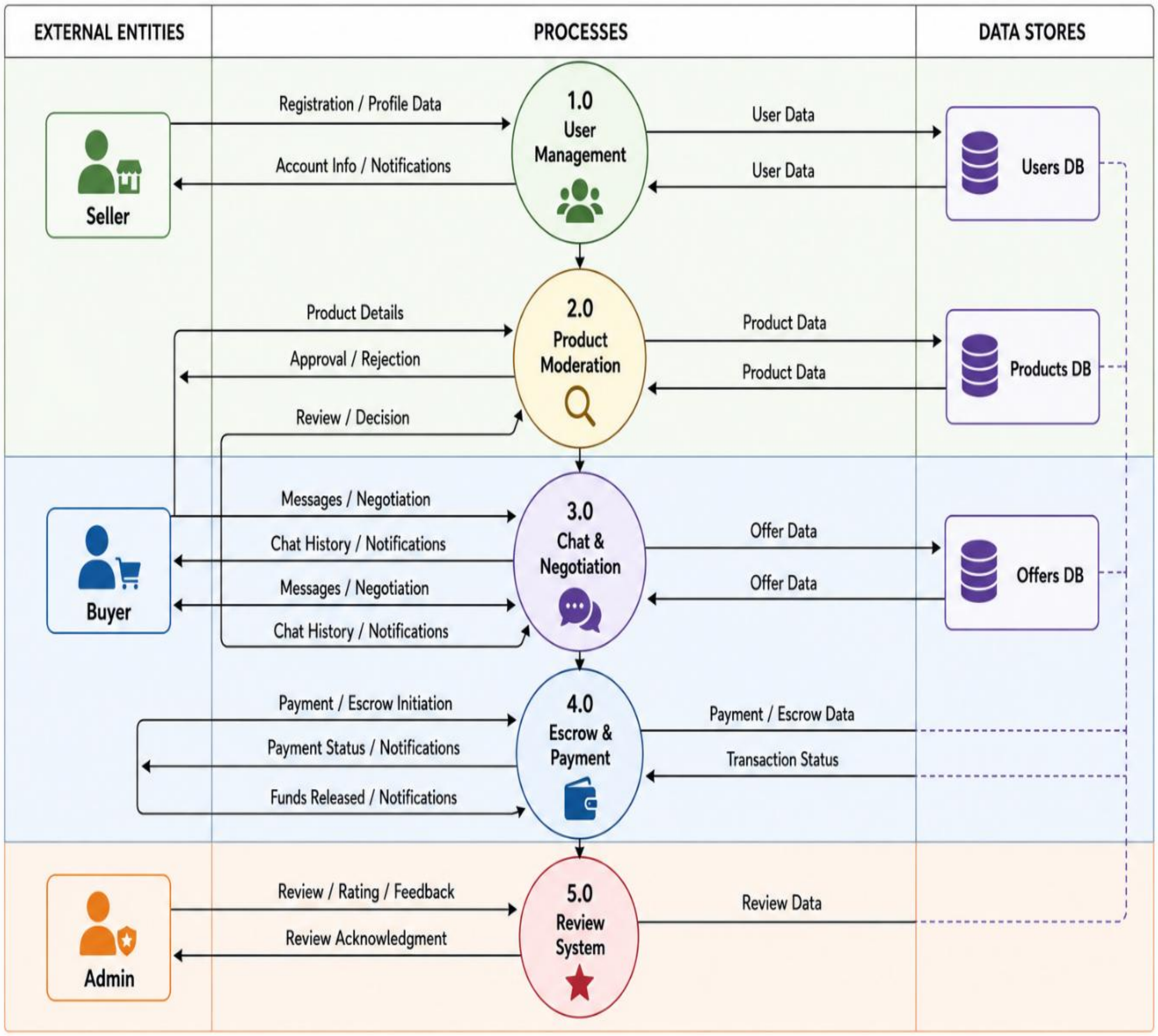


Figure 7.2 DFD Level 1 – Functional Decomposition

The Level 1 DFD is divided into five highest level functional modules of the system. It captures the flow of data between processes, data stores. Process (Product Moderation) is such a process that manages the flow into the Products DB of the listings, while Process (Escrow & Payment) organizes the safe movement of funds, interfacing with the Users DB for wallet balances and the Offers DB for transaction.

7.1.3 DFD Level 2: Detailed Escrow & Financial Flow

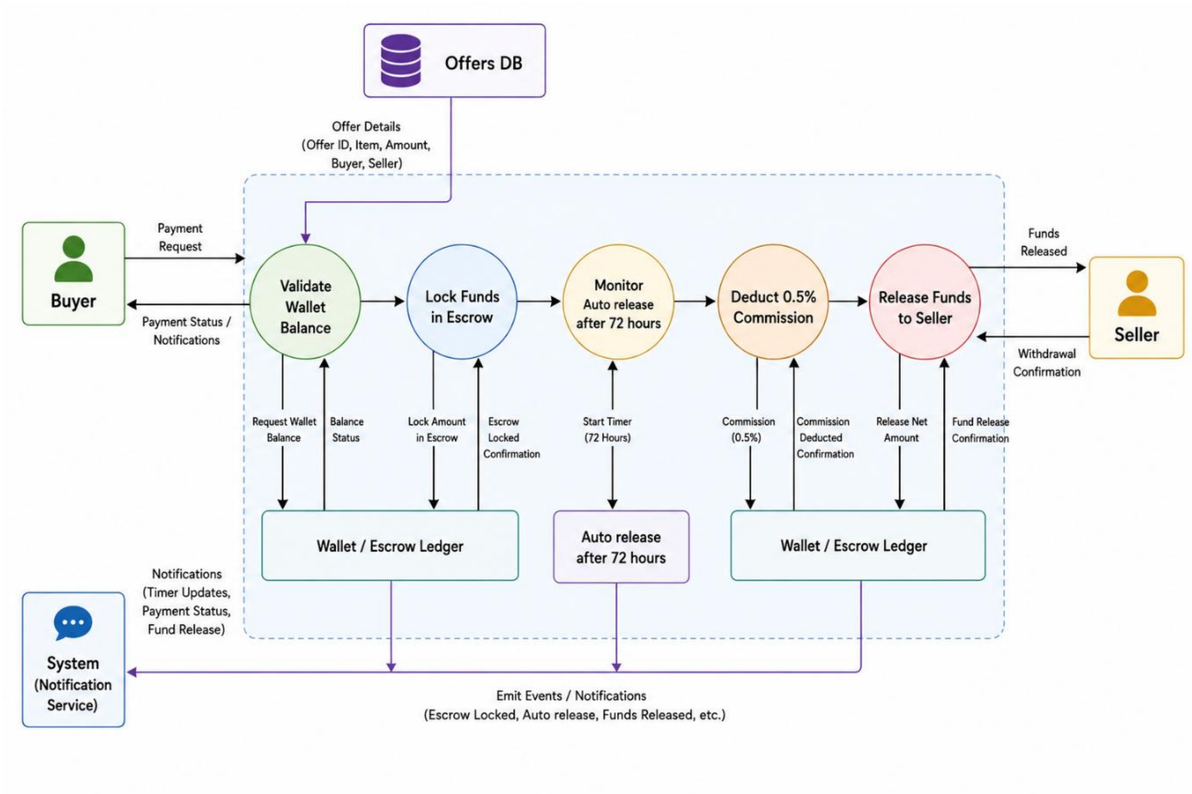


Figure 7.3 DFD Level 2 – Detailed Escrow & Financial Flow

The Level 2 DFD consists the detail logic of the core of the marketplace. It shows the process of data-based events to protect a transaction. The flowchart traces the flow of data from the beginning check balance to the critical state-based 72-hour timer and ends with the platform commission being automatically deducted before the Seller receives the funds.

7.2 Entity Relationship (ER) Diagram

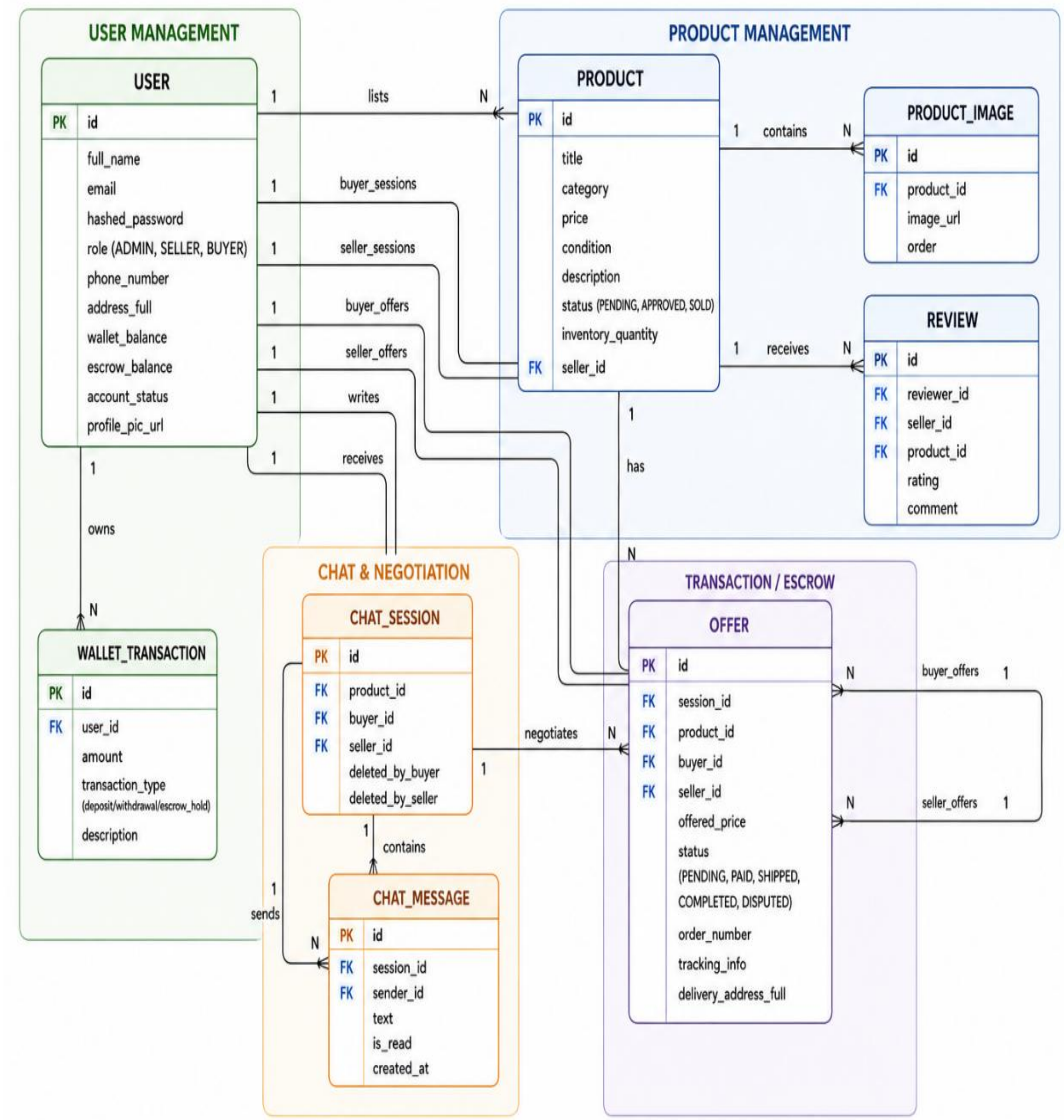


Figure 7.4 ER Diagram for ReSale Marketplace

7.3 Database Field Design

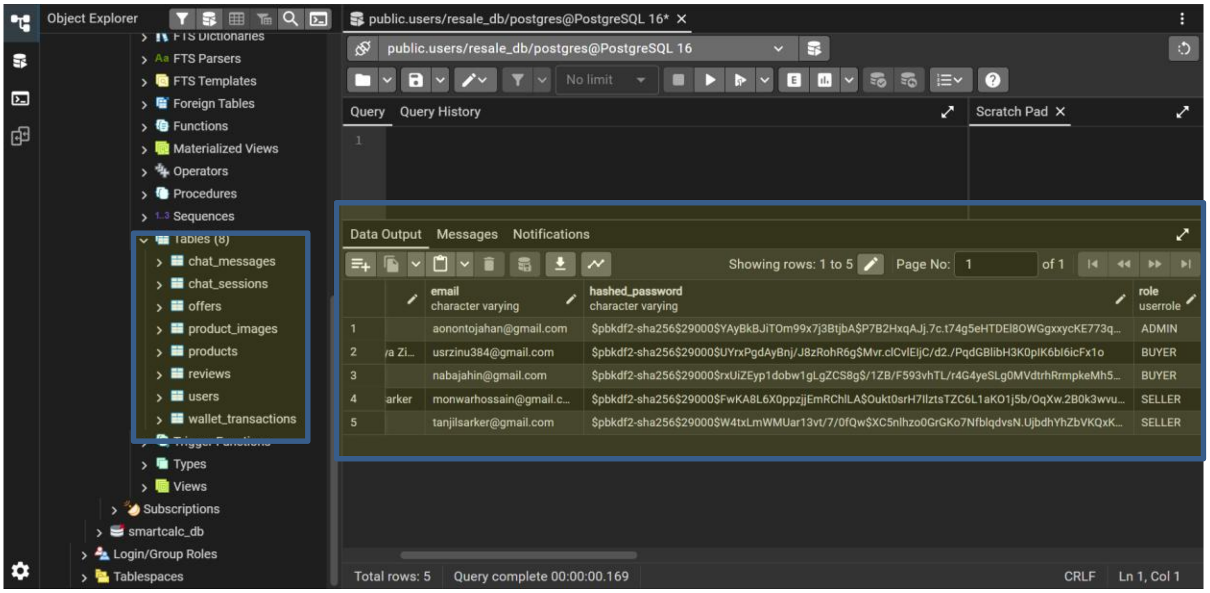


Figure 7.5 Database and User Table's Overview

7.4 Interface Design

The ReSale Marketplace system interface was designed to be simple and clean, and it allows both sellers and buyers to enjoy a seamless, tiptop experience. The design is simplistic, intuitive and easy to use, allowing users to quickly find products, manage listings and transactions.

The interface of the system includes the home page, user authentication page, product list page, escrow wallet page, order history page and user dashboard. Each of the interfaces were created with a clean, easily navigated layout and all have a consistent color scheme to enhance the user experience.

7.4.1 Home page

Figure 7.6 Home Page

7.4.2 Registration Page

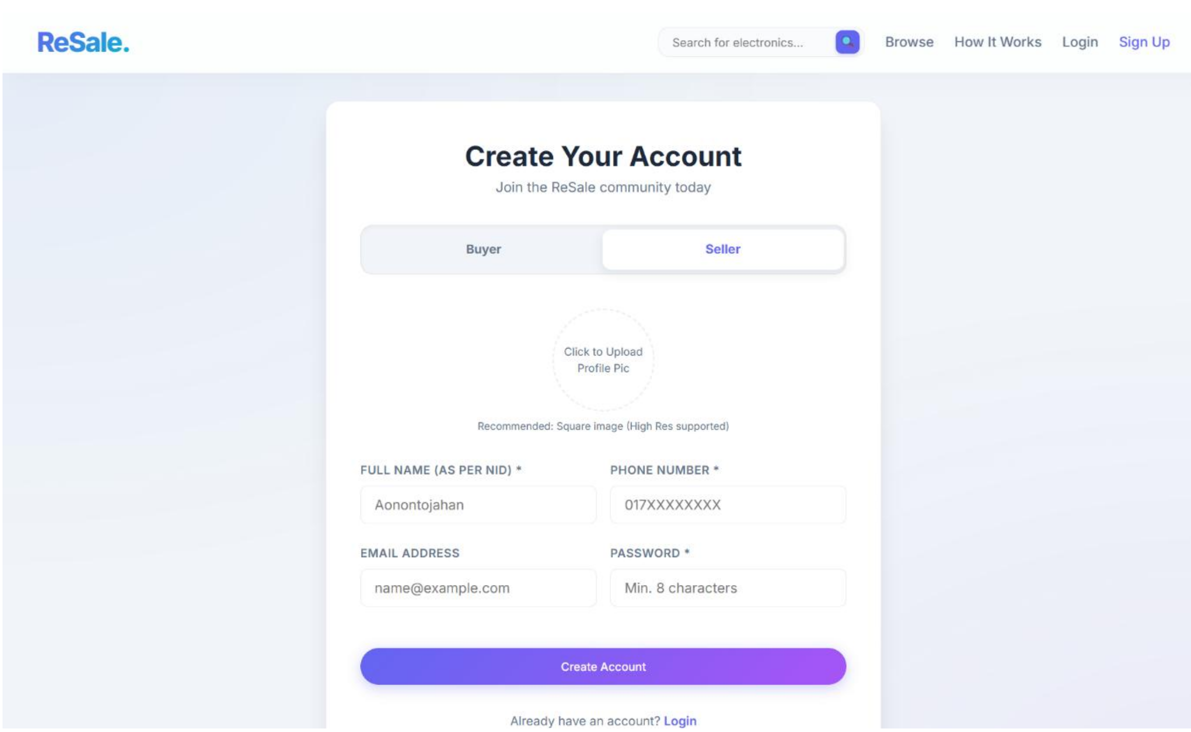


Figure 7.7 Seller View of the Registration Page

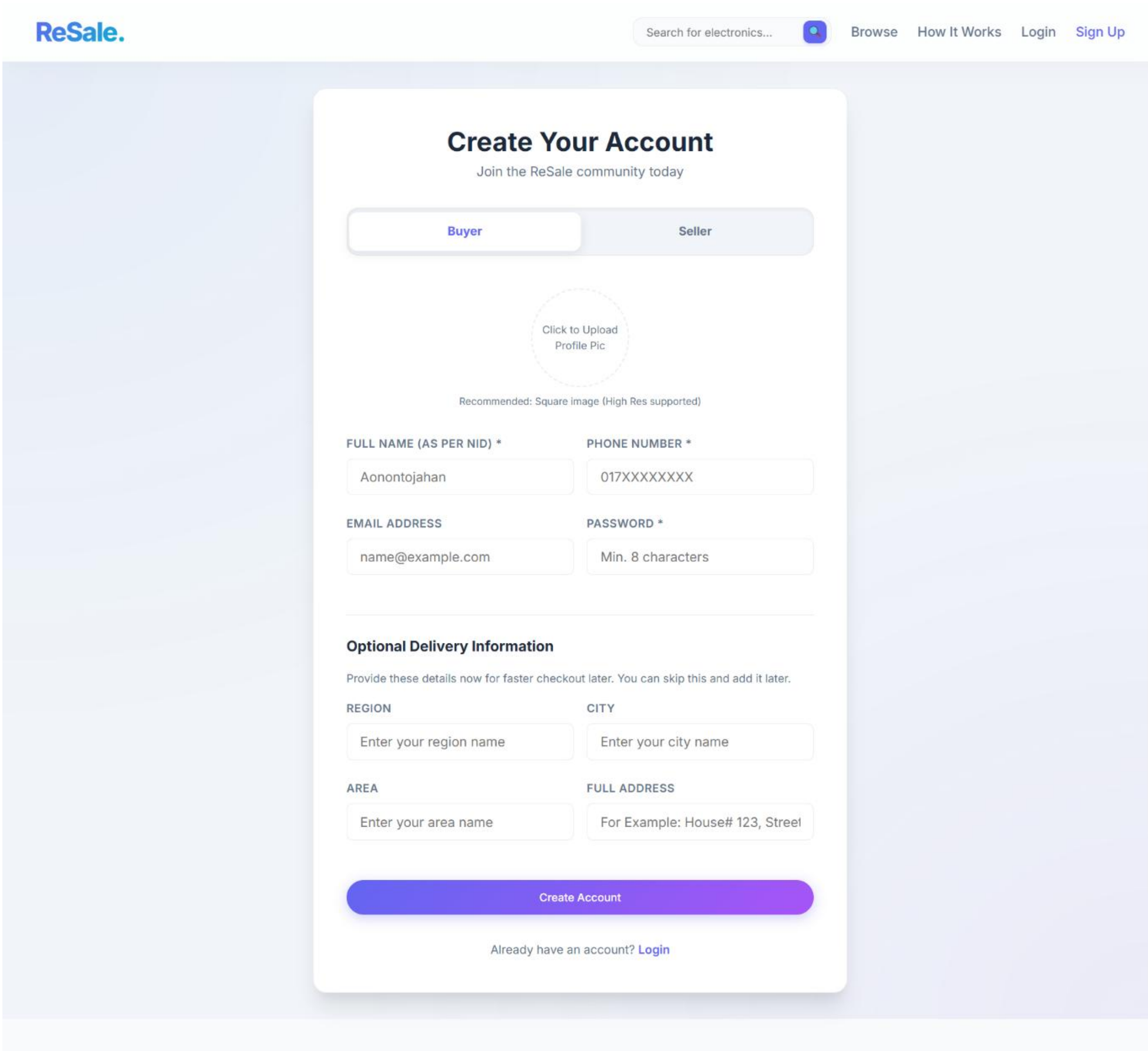


Figure 7.8 Buyer View of the Registration Page

7.4.3 Login Page

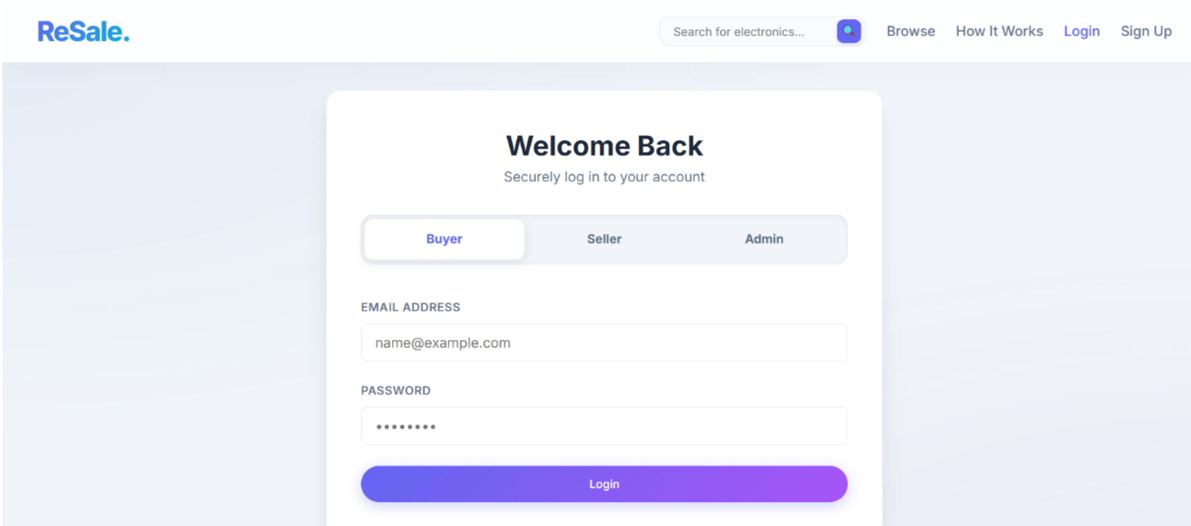


Figure 7.9 Login Page

7.4.4 Admin Interfaces

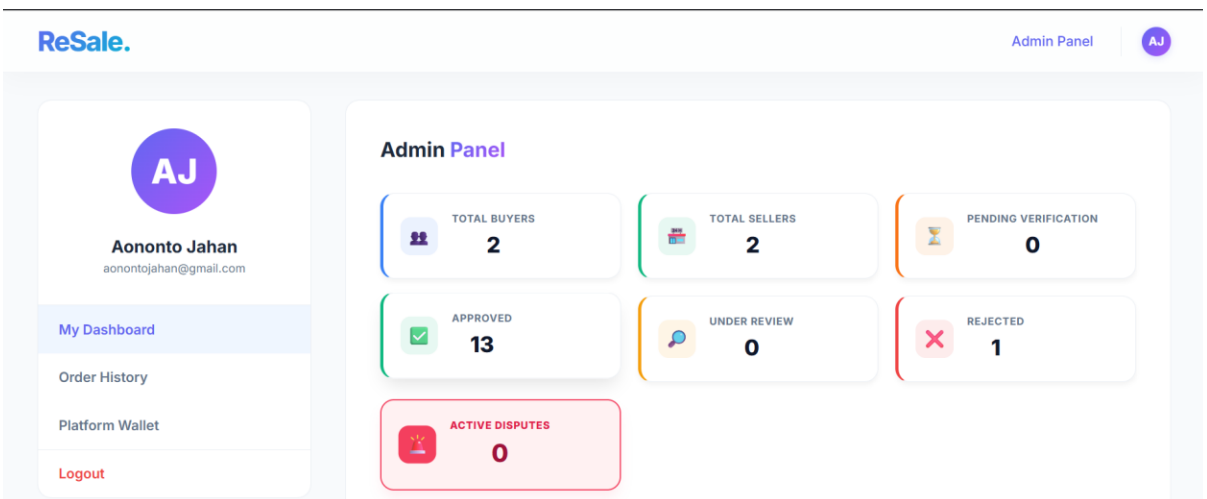


Figure 7.10 Admin Dashboard

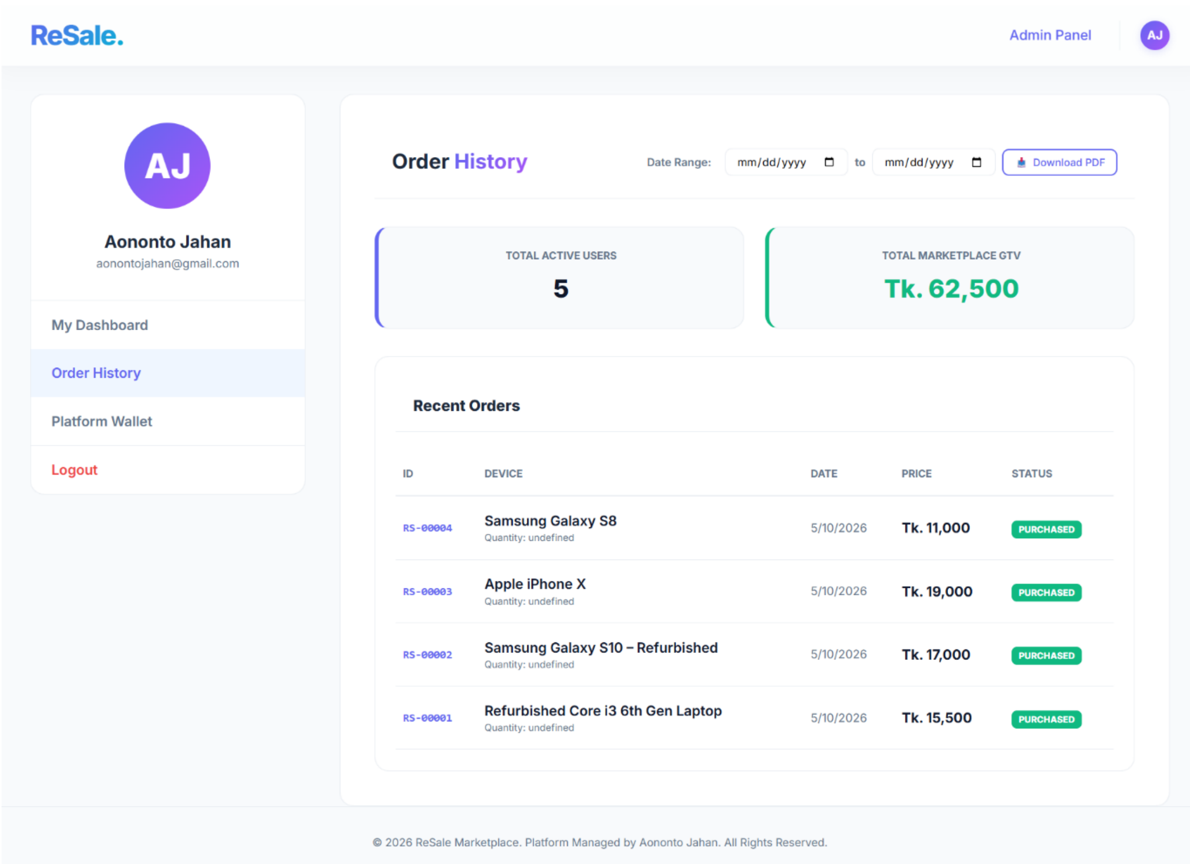


Figure 7.11 Platform Order History Page

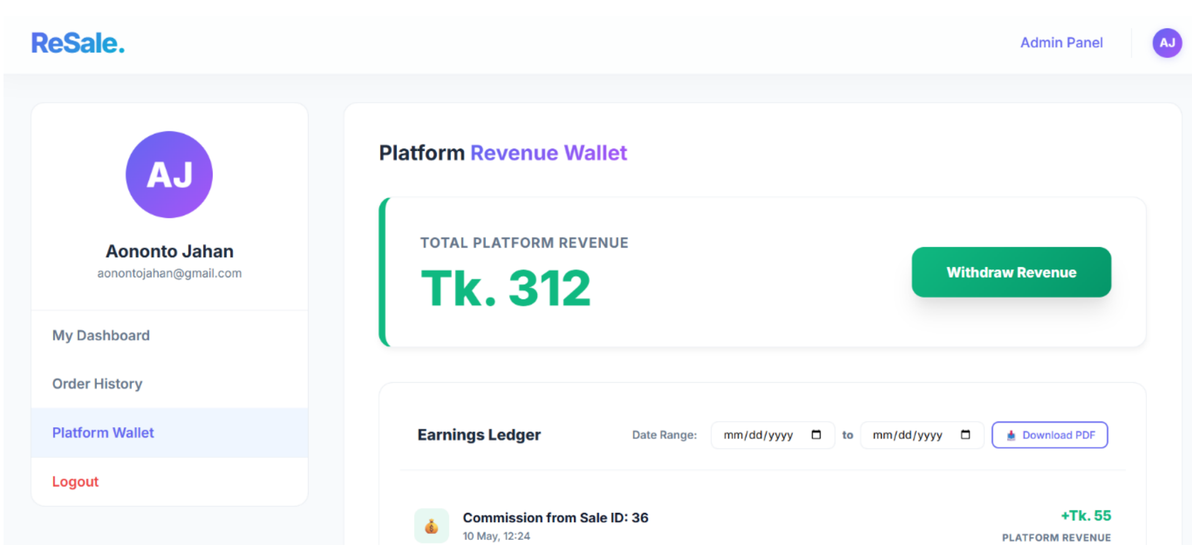


Figure 7.12 Platform Wallet Page

7.4.5 Buyer Interfaces

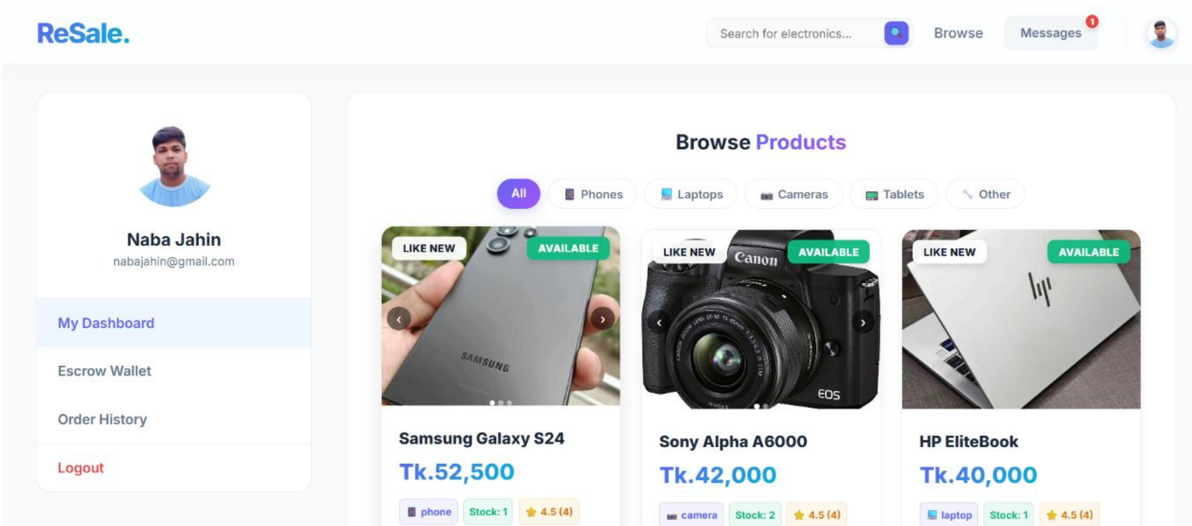


Figure 7.13 Buyer Dashboard Page

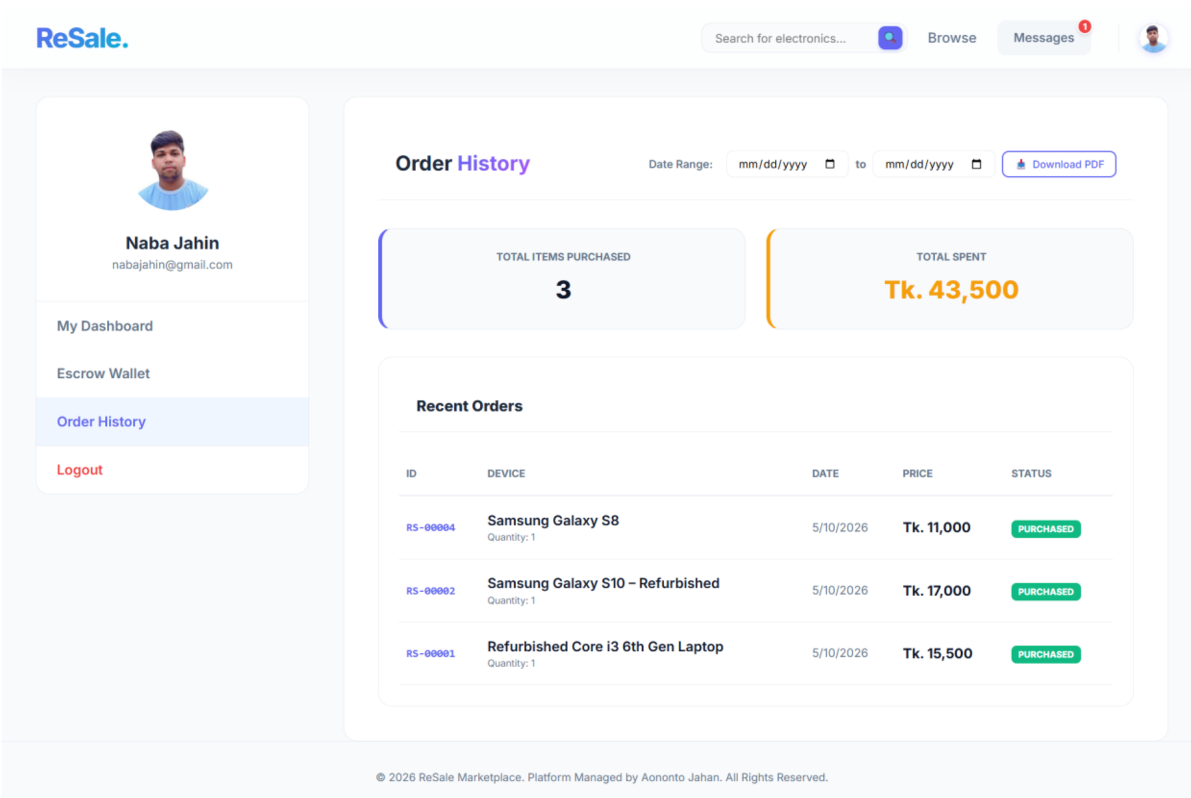


Figure 7.14 Buyer Order History Page

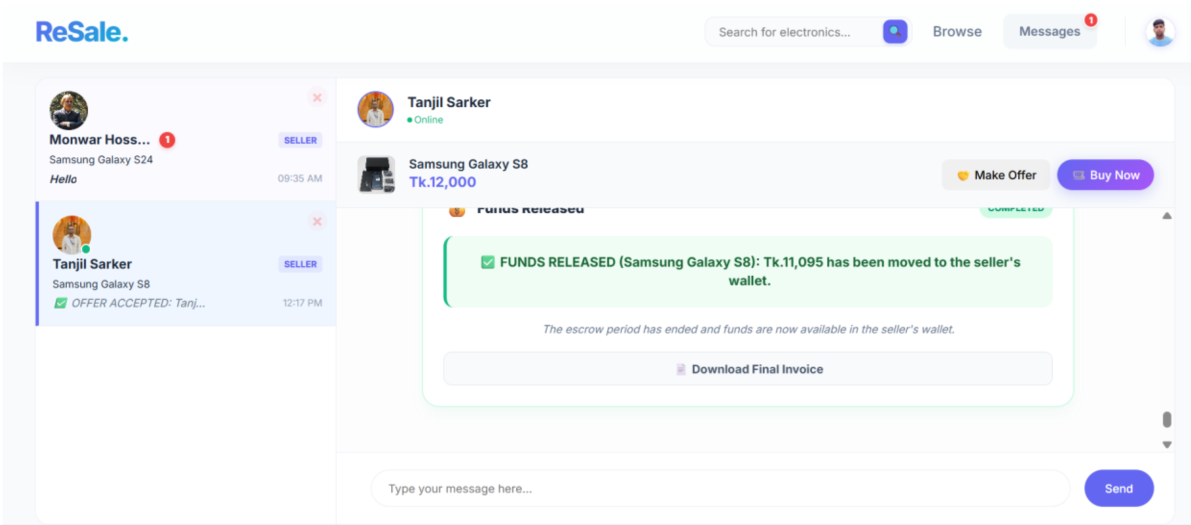


Figure 7.15 Buyer Negotiation Page

7.4.6 Seller Interfaces

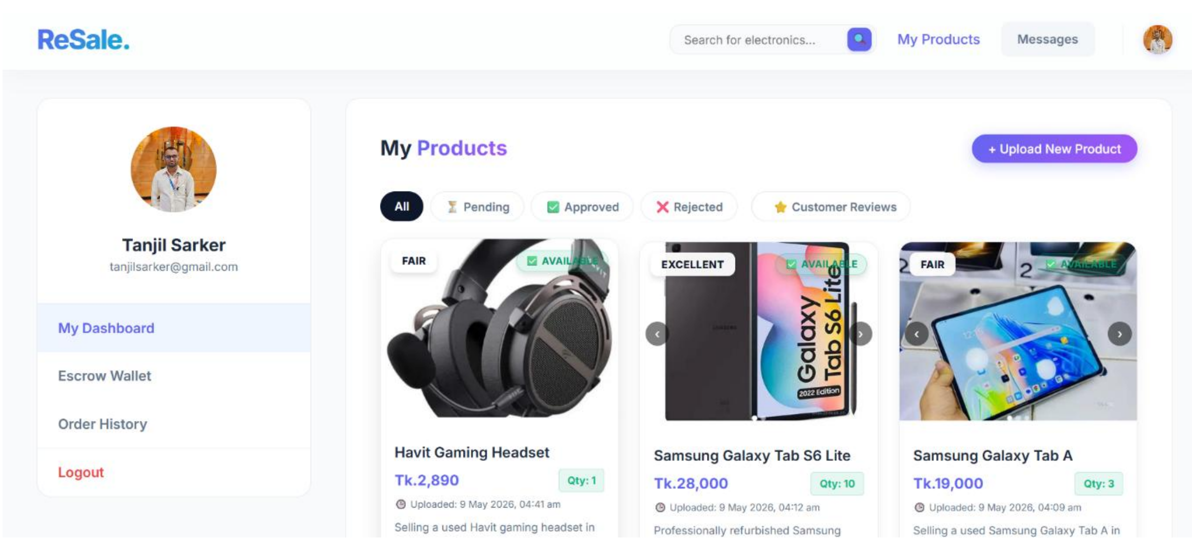


Figure 7.16 Seller Dashboard

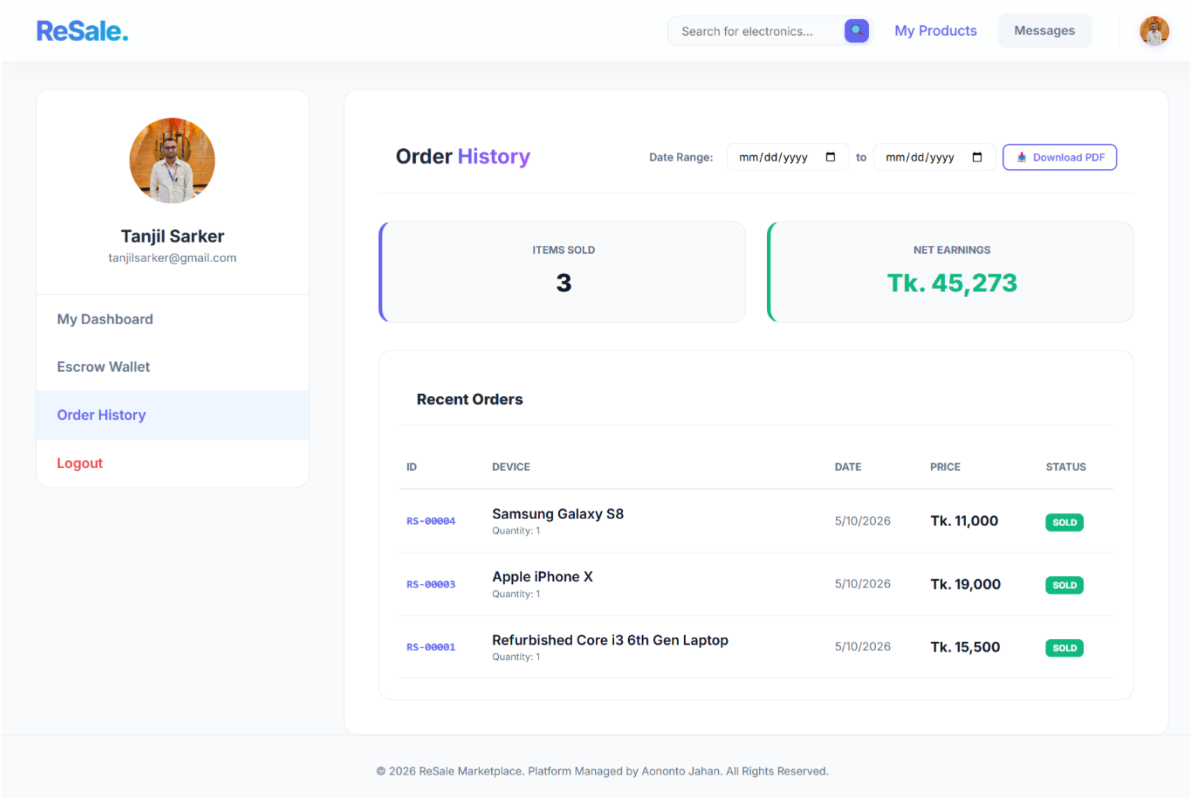


Figure 7.17 Seller Order History Page

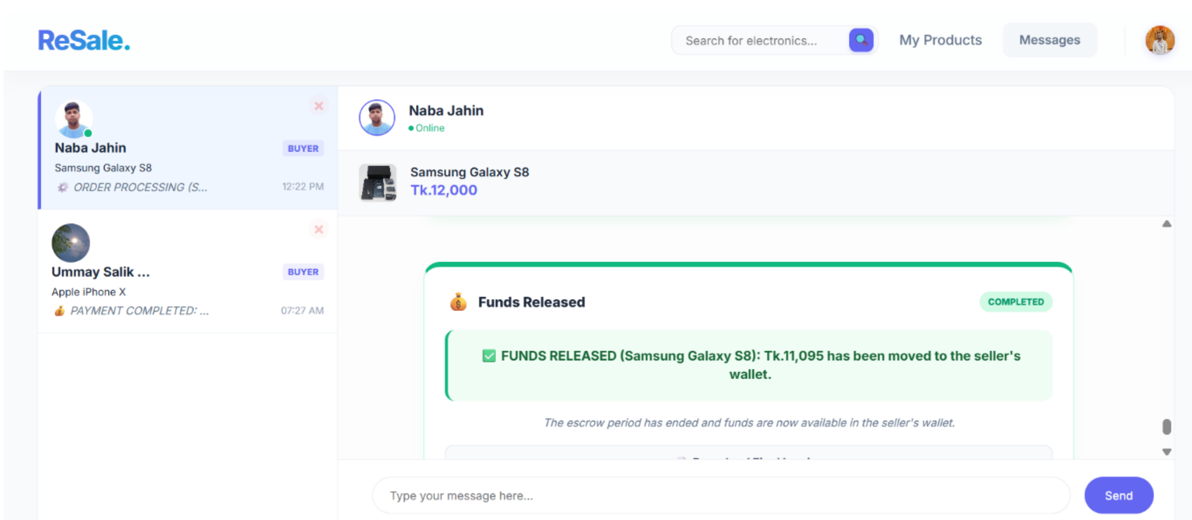
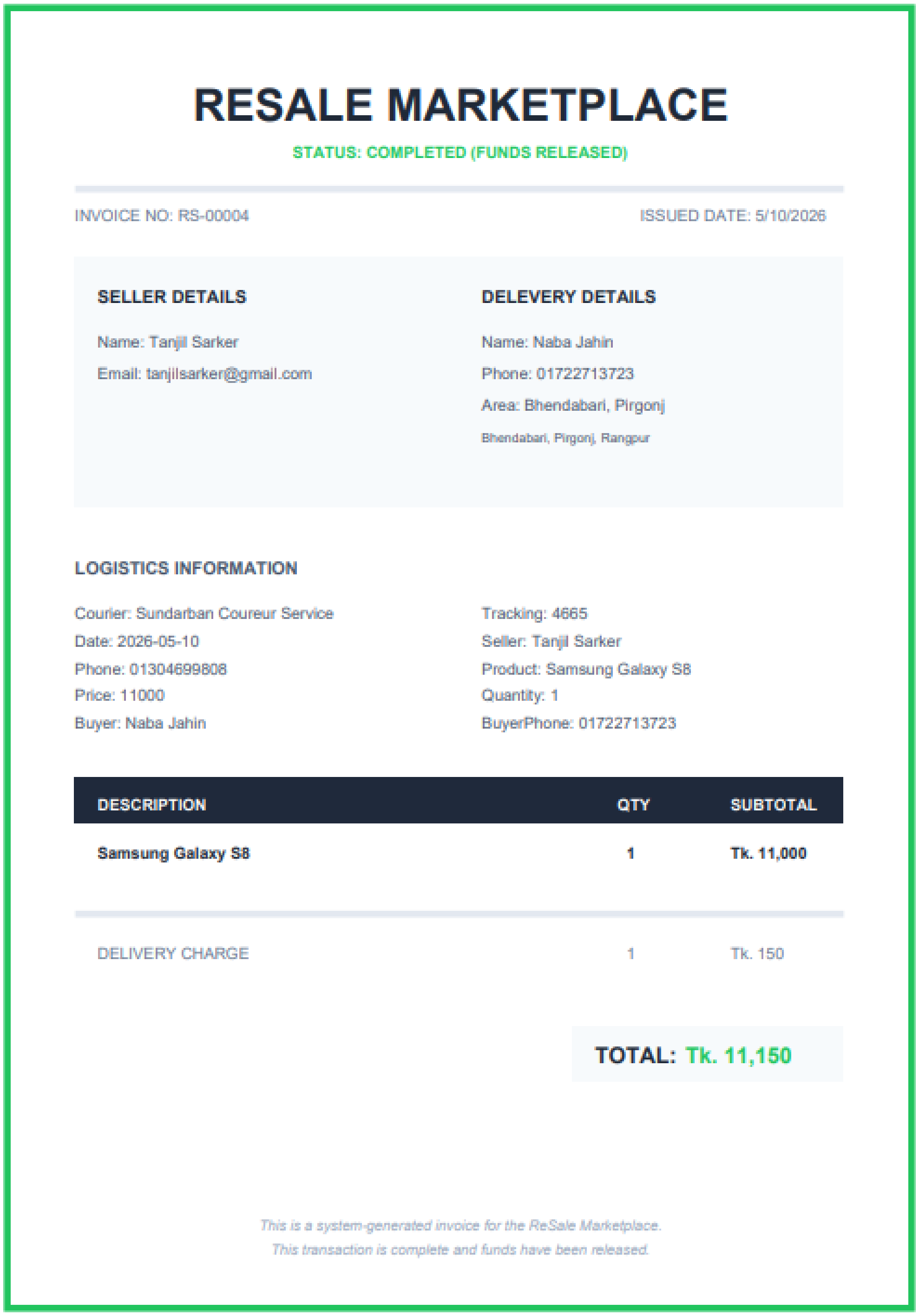


Figure 7.18 Seller Negotiation Page

7.4.7 Invoice



RESALE MARKETPLACE

STATUS: COMPLETED (FUNDS RELEASED)

INVOICE NO: RS-00004

ISSUED DATE: 5/10/2026

SELLER DETAILS

Name: Tanjil Sarker

Email: tanjilsarker@gmail.com

DELEVERY DETAILS

Name: Naba Jahin

Phone: 01722713723

Area: Bhendabari, Pirgonj

Bhendabari, Pirgonj, Rangpur

LOGISTICS INFORMATION

Courier: Sundarban Coureur Service

Date: 2026-05-10

Phone: 01304699808

Price: 11000

Buyer: Naba Jahin

Tracking: 4665

Seller: Tanjil Sarker

Product: Samsung Galaxy S8

Quantity: 1

BuyerPhone: 01722713723

DESCRIPTION	QTY	SUBTOTAL
Samsung Galaxy S8	**1**	**Tk. 11,000**
DELIVERY CHARGE	1	Tk. 150

TOTAL: Tk. 11,150

This is a system-generated invoice for the ReSale Marketplace.

This transaction is complete and funds have been released.

Figure 7.19 Invoice

Chapter 8.

Testing and Quality Assurance

8.1 Test Cases

A test case is a set of conditions or variables under which the tester knows whether a feature or function of an application is working correctly or not. These test cases are developed to check that the system fulfills the requirements, to find bugs, and to make the software dependable and working.

8.1.1 Test Case 1 (User Registration)

Table 8.1 Test Case 1

<table>
<tr><td colspan="3">Field</td><td colspan="3">Details</td></tr>
<tr><td colspan="3">Test Case ID</td><td colspan="3">1</td></tr>
<tr><td colspan="3">Test Priority</td><td colspan="3">High</td></tr>
<tr><td colspan="3">Module Name</td><td colspan="3">User Registration</td></tr>
<tr><td colspan="3">Test Title</td><td colspan="3">New User Signup Test</td></tr>
<tr><td colspan="3">Test By</td><td colspan="3">Aononto Jahan Junnurain</td></tr>
<tr><td colspan="3">Execution Steps:</td><td colspan="3"></td></tr>
<tr><td>Step</td><td>Test Steps</td><td>Test Data</td><td>Data accepted by form validation</td><td>Actual Result</td><td>Status</td></tr>
<tr><td>01</td><td>Fill Registration Form</td><td>Name:Naba Jahin,
Email:
nabajahin@gmail.com,
Password:nabajahin,
Role: Buyer</td><td>Data accepted by form validation</td><td>Data Accepted</td><td>Pass</td></tr>
<tr><td>02</td><td>Upload Profile Picture</td><td>image.jpg (500KB)</td><td>File uploaded and path saved</td><td>Success</td><td>Pass</td></tr>
</table>

03	Click "Register"		Account created; redirected to Home/Dashboard	Success	Pass

8.1.2 Test Case 2 (User Login)

Table 8.2 Test Case 2

Field			Details		
Test Case ID			2		
Test Priority			High		
Module Name			User Login		
Test Title			Role-Based Dashboard Access Test		
Test By			Aononto Jahan Junnurain		
Execution Steps:					
Step	Test Steps	Test Data	Expected Result	Actual Result	Status
01	Login as Admin	Email: aonontojahan@gmail.com, Pass: aonntojahan	Access to Admin Analytics & User Management	Admin Dashboard	Pass
02	Login as Seller	Email: tanjilsarker@gmail.com, Pass: tanjilsarker	Access to "My Products" & "Sales" panels	Seller Panel	Pass
03	Logout		Token cleared; redirected to Login	Redirected	Pass

8.1.3 Test Case 3 (Product Management)

Table 8.3 Test Case 3

Field			Details		
Test Case ID			3		
Test Priority			High		
Module Name			Product Management		
Test Title			Electronics Product Upload Test		
Test By			Aononto Jahan Junnurain		
Execution Steps:					
Step	Test Steps	Test Data	Expected Result	Actual Result	Status
01	Enter Product Info	Title: iPhone 13 Pro, Price: 85000, Category: Phones	Form fields filled correctly	Success	Pass
02	Upload 3 Images	Front, Back, and Side views	All 3 images stored and thumbnails shown	Saved	Pass
03	Submit Upload		Product appears as "Pending" in Seller list	Success	Pass

8.1.4 Test Case 4 (Chat & Negotiation)

Table 8.4 Test Case 4

Field			Details		
Test Case ID			4		
Test Priority			High		
Module Name			Chat & Negotiation		
Test Title			Real-time Offer & Acceptance Test		
Test By			Aononto Jahan Junnurain		
Execution Steps:					
Step	Test Steps	Test Data	Expected Result	Actual Result	Status
01	Make Offer (Buyer)	Offered Price: 80000	Offer card appears in chat for both parties	Card shown	Pass
02	Accept Offer (Seller)	Front, Back, and Side views	Status updates to "Accepted"; Pay button appears	Status: ACCEPTED	Pass
03	Send Message	Deal finalized!	Message delivers instantly via WebSocket	Delivered	Pass

8.1.5 Test Case 5 (Wallet & Escrow)

Table 8.5 Test Case 5

Field			Details		
Test Case ID			5		
Test Priority			High		
Module Name			Wallet & Escrow		
Test Title			Secure Escrow Payment Test		
Test By			Aononto Jahan Junnurain		
Execution Steps:					
Step	Test Steps	Test Data	Expected Result	Actual Result	Status
01	Click "Pay Now"	Offer Price + 150 Delivery Fee	Wallet deducted; Escrow balance increased	Balances updated	Pass
02	Verify Receipt		PDF Invoice generated with Order ID	PDF Downloaded	Pass
03	Check Status		Order status changes to "PAID"	Status: PAID	Pass

8.2 Test Summary Table

Table 8.6 Test Summary Table

Total Test Cases	Passed	Failed	Blocked	Pass Percentage
5	5	0	0	100%

Chapter 9

Ethical Consideration and Sustainability

9.1 Ethical Considerations

Basic values such as transparency, trust, and responsibility to the environment were the driving forces behind the development of the Resale Marketplace. It's all about building a secure, consumer-to-consumer environment for electronics. It is designed to win for buyers and sellers, by guaranteeing fair trade and data integrity the cornerstones of the marketplace's code of ethics.

9.1.1 Data Privacy and Security

The system processes sensitive information about users, such as their transaction history and personal contact information, with a high degree of security:

- **Authentication:** Secure authentication is provided by FastAPI with Bcrypt for password hashing and JWT (JSON Web Token) for stateless session management.
- **Authorization:** System architecture allows RBAC (Role Based Access Control) where Buyers are disallowed in Seller dash board, and normal users are disallowed in Admin's financial records and dispute logs.
- **Integrity:** Database transactions are employed to prevent from data damage at 72 hours escrow period, and wallet balance updates.
- **Privacy:** Buyers and sellers have their own private chat logs, which are encrypted and can only be viewed by the chat participants (and the Admin, should a dispute arise).

9.1.2. Intellectual Property

The project protects IP under terms of their respective MIT/Apache licenses by leveraging modern open source frameworks (FastAPI, Python, and PostgreSQL). All custom logic is owned by the developer, including the 72-hour Escrow Release mechanism, the automated Invoice Generation engine, and the manual negotiation protocol. There was no use of unlicensed or pirated 3rd party libraries in this implementation.

9.1.3. User Impact

And the very positive impact on the user by encouraging a Circular Economy was exported in the forefront. In this way, high-quality devices become financially accessible to a larger population through the resale of electronics. The platform's easy-to-use interface brings intricate procedures such as bargaining and payment into one consolidated application, eliminating the need for non-tech savvy users to have any technical knowledge.

9.1.4. Bias and fairness

Fairness is at the heart of this marketplace. The system is intended to serve equally.

- **Algorithmic Fairness:** Visibility of products and results in search are determined by neutral factors (price, condition, date) rather than any biased profiling.
- **Negotiation Fairness:** The system equalizes the negotiating process between buyers and sellers in the offer process, so that no side has an undue technological advantage.

9.1.5. Responsibility to Stakeholders

Prime stakeholders: Admin, Sellers and Buyers the system self is accountable by:

- **For the Admin:** Accurate revenue reports, and transparent and efficient dispute resolution tools to keep the platform healthy.
- **Sellers:** Safeguarding their interests, with payments taken in escrow before shipping.
- **Buyers:** Safeguarding their investment with a 72-hour inspection period prior to release of funds to the seller.

9.1.6. Professional Responsibility

As a developer, I was responsible to keep the code clean (PEP 8 for python) and the system stable and reliable. Good documentation and API schemas (Swagger) are kept so that the platform can be easily understood and scaled by future developers. We actively monitored and applied all security patches for dependencies.

9.2 Sustainability Through Software Engineering

Sustainability of the project is guaranteed by code optimization and through an efficient use of resources. Using asynchronous programming further allows the system to make full use of the server CPUs while keeping the CPU and memory consumption low and finally the carbon footprint of the hosting platform.

9.2.1. Economical Sustainability

The market operates with a scalable infrastructure that keeps the operational expenditure low.

- **Efficiency:** Due to lightweight backend and optimized frontend the system can run on cheap cloud infrastructure and still serve thousands of users.
- **Sustainable Expansion:** The modular approach makes it possible to introduce new functionality (for example crypto currency payments or AI-based price suggestions) without having to rewrite the entire system, safeguarding the original investment.

9.2.2. Social Sustainability

- **E-Waste Reduction:** our project is aligned with social and environmental sustainability as it helps to prolong the life of electronic components, which in turn minimizes the amount of e-waste generated worldwide and the need for new, resource-demanding manufacturing.
- **Community Trust:** With a secure Escrow process, the software creates digital trust within communities, fostering safer peer-to-peer commerce and contributing to the financial health of individual vendors.

Chapter 10.

Conclusion

10.1 Conclusion

The informal resale of electronics on social media or on unverified forums tends to be an issue of trust, with high risk of scams and unreliable communication. This Resale Marketplace provides a dependable, safe, and transparent system by utilizing a strong 72-hour escrow system and a negotiated module structure. Developed with FastAPI and SQL database, the platform facilitates seamless flow of information between Buyers, Sellers and Admins. In the end, the platform simplifies the process for peer-to-peer transactions and creates a secure environment that inspires consumers to reuse their electronics, helping build a more sustainable circular economy.

10.2 Limitations

Although the system is complete, the following constraints have been identified during implementation which are out of scope for this project:

- **Web Only Access:** The platform is currently a web application and does not have a native mobile application yet for real-time notifications.
- **Manually track shipment:** System manages escrow and payments, but does not currently connect to third-party courier APIs to fetch on the go real time shipping updates.
- **Testing payment gateway:** The existing wallet system runs on a mock balance. It is not yet integrated with a real banking or international payment gateway (like Stripe)
- **Partial Identity Verification:** Users trust each other based on reviews and transaction history, but the system currently does not require sellers to verify themselves using a formal NID or Passport.
- **Language Support:** The interface is now only in English and thus might be inaccessible to many locals in non-English speaking countries.

10.3 Future Plans

In order to increase the value and reach of the platform, the following are the envisaged enhancements to be made in the future development:

- **AI-Powered Price Suggestions**: Use ML (such as KNN in conjunction with Linear Regression) to recommend the "fair market value" for electronics given their condition and age while taking market trends into account.
- **Mobile Application Development**: Building the mobile application version in Flutter or React Native, including push notification feature for real-time chat and provide updates.
- **Shipping API Integration**: Partnering with our local logistics providers (Pathao, RedX or Fedex) to enable users to print shipping label and track packages from within the marketplace.
- **Live Payment Gateway:** Transitioning from mock wallets to live payment integration to enable users deposit/withdraw the real money through safe bank transfer or mobile financial solutions.
- **Multi-Language Support:** Incorporation of localization related features by adding the option for users to change the application language e.g English and Bengali to enable a wider user base.

References

Amigoscode (2021) FastAPI Tutorial - Building a REST API with Python. Available at: https://www.youtube.com/watch?v=GN6ICac3OXY (Accessed: 14 September 2025).

Ba, S. and Pavlou, P.A. (2002) 'Evidence of the effect of trust building technology in electronic markets: price premiums and buyer behavior', MIS Quarterly, 26(3), pp. 243–268.

Ellen MacArthur Foundation (2013) towards the circular economy: economic and business rationale for an accelerated transition. Available at: https://www.ellenmacarthurfoundation.org/publications (Accessed: 14 September 2025).

Ellen MacArthur Foundation (2017) Explaining the Circular Economy re-thinking progress. Available at: https://www.youtube.com/watch?v=zCRKvDyyHmI (Accessed: 14 September 2025).

Escrow.com (2018) How does Escrow work? Available at: https://www.youtube.com/watch?v=vVjH9H0F9p0 (Accessed: 14 September 2025).

FastAPI (2025) FastAPI Documentation: Concurrency and await. Available at: https://fastapi.tiangolo.com/async/ (Accessed: 14 September 2025).

Ghisellini, P., Cialani, C. and Ulgiati, S. (2016) 'A review on circular economy: the expected transition to a balanced interplay of environmental and economic systems', Journal of Cleaner Production, 114, pp. 11–32.

Kurose, J.F. and Ross, K.W. (2017) Computer networking: a top-down approach. 7th edn. Harlow: Pearson.

Laudon, K.C. and Laudon, J.P. (2020) Management information systems: managing the digital firm. 16th edn. Harlow: Pearson.

Lubanovic, B. (2019) Introducing Python: modern computing in simple packages. 2nd edn. Sebastopol, CA: O'Reilly Media.

National Geographic (2019) the global e-waste problem. Available at: https://www.youtube.com/watch?v=mDshS1iT8YQ (Accessed: 14 September 2025).

Nixon, R. (2021) Learning PHP, MySQL & JavaScript: with jQuery, CSS & HTML5. 6th edn. Sebastopol, CA: O'Reilly Media.

Silberschatz, A., Korth, H.F. and Sudarshan, S. (2019) Database system concepts. 7th edn. New York: McGraw-Hill.

The New Economy (2020) How the circular economy will change the world. Available at: https://www.youtube.com/watch?v=X6HJmG_5I-s (Accessed: 14 September 2025).

Tiangolo, S. (2020) FastAPI - A Python framework for high-performance APIs. Available at: https://www.youtube.com/watch?v=3DL7zG0-y-g (Accessed: 14 September 2025).

Traversy Media (2022) Real-time Chat with WebSockets and Node.js (Conceptual logic for Chat Apps). Available at: https://www.youtube.com/watch?v=jD7FnbI76Hg (Accessed: 14 September 2025).

Turban, E., Outland, J., King, D., Lee, J.K., Liang, T.P. and Turban, D.C. (2018) Electronic commerce 2018: a managerial and social networks perspective. 9th edn. Cham: Springer Texts in Business and Economics.